

PROJECT MANUAL

**REPLACEMENT OF FOGGING CAB GLASS
ADDISON AIRPORT
16000 DOOLEY ROAD
ADDISON, TX**



**FEDERAL AVATION ADMINISTRATION
10101 HILLWOOD PARKWAY
FORT WORTH, TEXAS 76177**

**TERMINAL ENGINEERING SERVICES
AJW-2C11C
FORT WORTH CENTRAL SERVICE AREA
FORT WORTH, TEXAS**

SECTION 00 01 10

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END OF SECTION 00 01 15

SECTION 01 00 00

GENERAL PARAGRAPHS

PART 1 - GENERAL

1.1 DRAWINGS, SPECIFICATIONS, AND OTHER CONTRACT DOCUMENTS

- A. Drawings showing general outlines and details necessary for a comprehensive understanding of the work form a part of the Contract Documents. The total number and the titles of the drawings constituting the Drawings are given in the index of the Drawings. All work under the Contract must be performed in all respects in compliance with the requirements of the Contract Documents.
- B. The Contract Documents provide for a complete work and may have been prepared in divisions of various crafts, trades and other categories of work. The Contractor is responsible for the performance of all work under the Contract regardless of any such divisions and must ensure that all of the work is performed and completed.
- C. The FAA will provide the Contractor with one bound copy of the construction drawings and specifications for the Contractor's use during the execution of the Contract. The Contractor may reproduce these documents for its use during the performance of the work under this Contract.
- D. The Contractor must maintain at the Site at all times at least one (1) copy of Drawings, Specifications and all other Contract Documents, together with at least one (1) complete set of approved Shop Drawings and approved samples.
- E. The Contract, Drawings, Specifications, and all referenced standards cited are essential parts of the Contract requirements. A requirement occurring in one is as binding as though occurring in all. They are intended to be complementary and to describe and provide for a complete work.
- F. On the drawings, calculated or figured dimensions govern over scaled dimensions.
- G. Generally the electrical drawings and any drawings relating to piping are schematic only and dimensions shall be followed without regard to scale. Drawings show the general arrangement and the extent of the work. Exact location and equipment arrangement shall be determined by the physical dimensions of the equipment actually furnished, by shop drawings, detailed dimensioned drawings. Submit all necessary shop drawings mentioned in drawings or specifications.
- H. In the event of a conflict between commercially available or industry standard documents and specific requirements in FAA Orders and Notices, the more stringent shall apply.
- I. The contractor is responsible to provide a certification to FAA that all materials used on this project are free of lead and asbestos
- J. The Contractor must not take advantage of any apparent error, omission, discrepancy, or ambiguity on the Drawings or Specifications. If any error, omission, discrepancy, or ambiguity is found by the Contractor in the Drawings or Specifications, the Contractor must refer these to the

Contracting Officer (CO) prior to beginning work on affected task(s), for interpretation and decision, and such decision must be final.

- K. The CO has the right to correct apparent errors or omissions in the Drawings and Specifications and to make such interpretations as he may deem necessary for the proper fulfillment of the Contract Documents. During the course of the work, should any conflicts, ambiguities, or discrepancies be found that are not addressed or any discrepancies between the Drawings and the Specifications to which the Contractor has failed to call attention before submitting the offer, then the CO will interpret the intent of the Drawings and Specifications and the Contractor hereby agrees to abide by the CO's interpretation and agrees to carry out the work in accordance with the decision of the CO. In such event the Contractor will be held to have included in the offer the most proper material and/or method of construction in order to fulfill the intent of the Contract Documents.
- L. When a material, article, or equipment is designated by a brand name, and more than one brand name is listed, it will be understood that the design is based on one of the brand name listed products. The Contractor will be responsible for all coordination necessary to accommodate the material, article, or equipment actually being provided and per the requirements of Section 01 25 00 without additional cost to the government.
- M. The organization of the specification into divisions, sections and articles, and the arrangement of Drawings does not restrict or limit the Contractor in dividing the Work among Subcontractors or in establishing the extent of work to be performed by any trade.
- N. Product and Reference Standards:
 - 1. When descriptive catalog designations including manufacturer's name, product brand name, or model number are referred to in the Contract Documents, such designations are considered as being those found in industry publications of current issue on the date of the first advertisement for offers.
 - 2. When standards of the Federal Government, State Department of Transportation, Standards Organization such as ASTM, AASHTO, AWS, or ANSI, trade societies, or trade associations are referred in the Contract Documents by specific date of issue, these are considered a part of this Contract. When such references do not bear a date of issue, the current published edition on the date of the first advertisement for offers are considered as part of the Contract.
 - 3. Where in the Contract Documents an item is identified by a particular manufacturer's name, model or other code it must be interpreted to include other manufacturers' product of like and equal quality whether the words "or equal" are included or not unless specifically stated otherwise.
 - 4. Wherever a particular manufacturer's product is required, to the exclusion of all others, appropriate language is included in the Contract Documents.
 - 5. Wherever the terms, "as directed", "ordered", "permitted", "designate", "as approved", "approved equal", "or equal", "acceptable" and other words of similar meaning which authorize an exercise of judgement are used in the Contract Documents, such judgment is vested only in the CO or designated representative.
 - 6. When a particular manufacturer's product is used, the Basis of Design and Section 01 25 00 shall be follow and be in conformance.

1.2 CONFORMITY WITH DRAWINGS AND SPECIFICATIONS

- A. No deviation from the Drawings, Specifications and other Contract Documents will be permitted without the prior written approval of the CO.

1.3 SUPERVISION AND CONSTRUCTION PROCEDURES

- A. At all times during performance of this contract, and until the work is completed and accepted, the Contractor must directly superintend the work or assign and have on the worksite a competent superintendent who is an employee of the Contractor and is satisfactory to the CO and has the authority to act for the Contractor.
- B. The Contractor must supervise and direct the Work, using the Contractor's best skill and attention. The Contractor is solely responsible for and has control over construction means, methods, techniques, sequences, and procedures and for coordinating all portions of the Work under the Contract including coordination of the duties of all trades, unless the Contract Documents give other specific instructions concerning these matters.
- C. The Contractor must control its operations and those of its Subcontractors and Suppliers to assure the least inconvenience to the traveling public. Under all circumstances, safety must be the most important consideration.
- D. Contractor must lay out all work well enough in advance to avoid conflicts or interferences with other work in progress so that in case of interference the layout may be altered to suit the conditions, prior to the installation of any work and without additional cost to the FAA. The Contractor must be responsible to coordinate all work and take all action as required to avoid conflicts between trades.
- E. The Contractor should make sure that all contractor and subcontractor personnel watch the FAA ORM for Contractors video prior to being on site.
 - 1. MyFAA TV: ORM for FAA Contractors: <https://employees.faa.gov/tv/?mediaId=2164>

1.4 CORRESPONDENCE

- A. Contract correspondence must be directed to the CO with a copy to the COR.
- B. Send submittals directly to the COR with a copy of the transmittal letter to the CO.

1.5 LIST OF SUBCONTRACTORS

- A. The Contractor must, within 10 calendar days after award, furnish to the CO with a copy to the COR, a list of subcontractors showing the type of work each will perform. If all subcontracts have not been awarded when the initial list is submitted, the Contractor must update the list monthly.

1.6 EXISTING WORK

- A. The disassembling, disconnecting, cutting, removal, or altering in any way of existing work must be carried on in such a manner as to prevent injury or damage to all portions of existing work, whether they are to remain in place, be re-used in the new work, or be salvaged and stored.

- B. All portions of existing work which have been cut, damaged, or altered in any way during construction operations must be repaired or replaced in kind in an approved manner to match existing or adjoining work. All work of this nature must be performed by the Contractor at his expense and must be as directed. Existing work must, at the completion of all operations, be left in a condition as good as existed before the new work started.

1.7 MATERIALS AND EQUIPMENT TO BE SALVAGED

- A. Except where specifically specified otherwise herein, or designated on the drawings, all existing materials and equipment which are required to be removed or disconnected to perform the work but are not indicated or specified for use in the new work, becomes the property of the Contractor and must be disposed of properly.

1.8 UNCOVERING AND CORRECTION OF WORK

A. Uncovering Work

- 1. If any portion of the Work is covered contrary to the COR's request or to requirements specifically expressed in the Contract Documents, it must, if required in writing by the COR be uncovered for his observation and be recovered (if corrections are not required) or be corrected, if applicable, at the Contractor's expense without change in the Contract Time.

B. Correction of Work

- 1. The Contractor must promptly correct Work rejected by the COR that fails to conform to the requirements of the Contract Documents, whether observed before or after Substantial Completion and whether or not fabricated, installed or completed. The Contractor must bear all costs of correcting such rejected Work, including additional testing and inspections and compensation for the COR's services and expenses incurred by the FAA.
- 2. If, within two years after the date of Substantial Completion of the Work or designated portion thereof, or after the date for commencement of warranties established above, or by terms of an applicable special warranty required by the Contract Documents, any of the work is found to be not in accordance with the requirements of the Contract Documents, the Contractor must correct it promptly after receipt of written notice from the FAA to do so unless the FAA has previously given the Contractor a written acceptance of that specific condition. This period of two years must be extended with respect to portions of Work first performed after Substantial Completion by the period of time between Substantial Completion and the actual performance of the Work. This obligation survives acceptance of the Work under the Contract and termination of the Contract. The FAA must give such notice within a reasonable amount of time after discovery of the condition.
- 3. The Contractor must remove from the site portions of the Work that are not in accordance with the requirements of the Contract Documents and are neither corrected by the Contractor nor accepted by the FAA.
- 4. If the Contractor fails to correct nonconforming Work within a reasonable time, the FAA may correct it in accordance with General Provisions. If the Contractor does not proceed with correction of such nonconforming Work within a reasonable time fixed by written notice from the COR, the FAA may correct or remove such nonconforming work and all costs for such corrections or removals must be assessed against the Contractor.
- 5. The Contractor must bear the cost of correcting destroyed or damaged Work, whether completed or partially completed, of the FAA or separate contractors caused by the

Contractor's performing correction or removal of Work which is not in accordance with the requirements of the Contract Documents.

6. Nothing contained herein shall be construed to establish a period of limitation with respect to other obligations that the Contractor might have under the Contract Documents. Establishment of the time period of two years as described above relates only to the specific obligation of the Contractor to correct the Work, and has no relationship to the time within which the obligation to comply with the Contract Documents may be sought to be enforced, nor to the time within which proceedings may be commenced to establish the Contractor's liability and damages with respect to the Contractor's obligations other than specifically to correct the Work.

1.9 LOCATION OF SERVICES

- A. The FAA does not guarantee the accuracy or the completeness of the location information relating to existing utility services, facilities, or structures that may be shown on the drawings. Any inaccuracy or omission in such information must not relieve Contractor of its responsibility to protect such existing features from damage or unscheduled interruption of service.

1.10 COOPERATE WITH OTHER ENTITIES

- A. Cooperate with the FAA and other public or private utility services, or a utility service of another government agency that may be authorized by the FAA to construct, reconstruct, or maintain such utility services or facilities during the progress of the work. Control operations to prevent the unscheduled interruption of such utility services and facilities.

1.11 DAMAGE TO SERVICES

- A. Should Contractor damage or interrupt the operation of a utility service or facility by accident or otherwise, it must immediately notify in writing the FAA/operator, appropriate public safety authorities, and the COR and must take all reasonable measures to prevent further damage or interruption of service. Cooperate with the utility service or facility FAA and the COR continuously until such damage has been repaired and service restored.

1.12 FAILURE TO PROTECT PROPERTY

- A. Contractor will not be entitled to any extension of time or compensation on account of Contractor's failure to protect all facilities, equipment, materials and other property as described herein. All costs in connection with any Improvements or restoration necessary or required by reason of unauthorized obstruction, damage, or use must be borne by Contractor.

1.13 UTILITY CONTRACTOR LICENSING REQUIREMENTS

- A. Contractor must comply with all state and local requirements for construction of utilities.

1.14 ASBESTOS AND LEAD FREE CERTIFICATION

- A. FAA policy is to construct all new facilities without asbestos or lead containing products. The Contractor must provide a letter on his company's standard letter head stating that to the best of his knowledge no product or material used on this project contains asbestos or lead. The statement must include the name of the project and the contract number and must be signed by an officer of

the company. The statement must be furnished within 10 calendar days of the Substantial Completion date. Submission of this statement is a condition for final payment under the contract.

- B. Verification: If the FAA suspects the presence of asbestos or lead, tests will be performed on the material or product at the FAA's expense. If it is determined that the product or material does contain asbestos or lead, then the Contractor must remove the product or material and replace at his own expense including the expense of the testing and any retesting that may be necessary.
- C. Non – Compliance: If the Contractor fails to provide the above statement, then the FAA may have a complete building survey performed by a qualified testing firm and the costs deducted from the Contractor's final payment.

1.15 SAFETY DATA SHEETS (SDS):

- A. The Contractor must submit to the COR Safety Data Sheets (SDS) for all materials and/or products utilized during the course of the project accomplishment. During the course of the project, both the COR and the Contractor must routinely check products utilized on-site to ensure only products which have had SDS submitted are utilized. Copies of all SDS must be turned over to the local FAA office for their records.

1.16 INITIAL SUBMITTALS

- A. The following submittals are required to be sent to the FAA fourteen (14) days after award.
 - 1. Section 01 10 00 SUMMARY OF WORK
 - a. Airspace Study (due to FAA within 5 calendar days of award)
 - 1) Proposed location for Equipment (Crane/ Lift)
 - 2) Specifications of Equipment (Crane/ Lift)
 - 2. Section 01 00 00 LIST OF SUBS, CERTIFICATE OF INSURANCE
 - 3. Section 01 32 00 CONSTRUCTION SCHEDULE
 - a. Preliminary
 - 1) Should include all submittals
 - 2) Should include all major item lead times
 - 3) Should include construction activities
 - 4. Section 01 40 00 CONTRACTOR QUALITY CONTROL
 - 5. Section 08 88 58 ATCT Cab Glass
 - 6. Traffic Control Plans
- B. All submittals listed above in letter A must be approved prior to a Notice to Proceed (NTP) is issued.
- C. All submittals submitted after NTP are subjected to the same submittal procedures in section 01 33 00 and is not a justification for delay and no time extensions will be given due to product lead times from these submittals.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 01 00 00

SECTION 01 10 00

SUMMARY OF WORK

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Work covered by the Contract Documents.
 - 2. Type of Contract.
 - 3. Work phases.
 - 4. Work under other contracts.
 - 5. Products ordered in advance.
 - 6. FAA-furnished products.
 - 7. Use of premises.
 - 8. FAA's occupancy requirements.
 - 9. Work restrictions.
 - 10. Specification formats and conventions.
 - 11. Permits and Fees
 - 12. Insurance

1.2 DEFINITIONS

- A. Contracting Officer's Representative (COR): Individual authorized to receive and distribute information on behalf of the Contracting Officer. Also referred to as the Contracting Officer's Representative (COR) and/or Resident Engineer (RE) in some instances.

1.3 INITIAL SUBMITTALS

- A. The following submittals are required to have FAA approval prior to Notice to Proceed.
 - a. Section 01 10 00 SUMMARY OF WORK: List of subs, certificate of insurance
 - b. Section 01 32 00 PROJECT SCHEDULE
 - c. Section 01 40 00 CONTRACTOR QUALITY CONTROL for CQC Plan
 - d. Section 01 71 33 PROTECTION OF WORK AND PROPERTY
 - e. Section 08 88 58 ATCT Cab Glass
 - f. Traffic Control Plans

1.4 CONFORMITY WITH DRAWINGS AND SPECIFICATIONS

- A. No deviation from the Drawings, Specifications, or any other Contract Documents will be permitted without the prior written approval of the CO.

1.5 DRAWINGS, SPECIFICATIONS, AND OTHER CONTRACT DOCUMENTS

- A. Drawings showing general outlines and details necessary for a comprehensive understanding of the work form a part of the Contract Documents. The total number and the titles of the drawings

constituting the Drawings are given in the index of the Drawings. All work under the Contract must be performed in all respects in compliance with the requirements of the Contract Documents.

- B. The Contract Documents provide for a complete work and may have been prepared in divisions of various crafts, trades and other categories of work. The Contractor is responsible for the performance of all work under the Contract regardless of any such divisions and must ensure that all of the work is performed and completed.
- C. The FAA will provide the Contractor with electronic copies of the construction drawings and specifications for the Contractor's use during the execution of the Contract.
- D. The Contractor must maintain at the Site at all times at least one (1) copy of Drawings, Specifications and all other Contract Documents, together with at least one (1) complete set of approved Shop Drawings and approved samples.
- E. The Contract, Drawings, Specifications, and all referenced standards cited are essential parts of the Contract requirements. A requirement occurring in one is as binding as though occurring in all. They are intended to be complementary and to describe and provide for complete work.
- F. On the drawings, calculated or figured dimensions govern over scaled dimensions.
- G. In the event of a conflict between commercially available or industry standard documents and specific requirements in FAA Orders and Notices, comply with CO's direction.
- H. The Contractor is responsible to provide a certification to FAA that all materials used on this project are free of lead and asbestos
- I. When a material, article, or equipment is designated by a brand name, and more than one brand name is listed, it will be understood that the design is based on one of the brand name listed products. The Contractor will be responsible for all coordination necessary to accommodate the material, article, or equipment actually being provided and per the requirements of Section 01 25 00 Substitution Procedures without additional cost to the government.
- J. The organization of the specifications into divisions, sections and articles, and the arrangement of Drawings does not restrict or limit the Contractor in dividing the Work among Subcontractors or in establishing the extent of work to be performed by any trade.
- K. Product and Reference Standards:
 - 1. When descriptive catalog designations including manufacturer's name, product brand name, or model number are referred to in the Contract Documents, such designations are considered as being those found in industry publications of current issue on the date of the first advertisement for offers.
 - 2. When standards of the Federal Government, State Department of Transportation, Standards Organization such as ASTM, AASHTO, AWS, or ANSI, trade societies, or trade associations are referred to in the Contract Documents by specific date of issue, these are considered a part of this Contract. When such references do not bear a date of issue, the current published edition on the date of the first advertisement for offers are considered as part of the Contract.
 - 3. Where in the Contract Documents an item is identified by a particular manufacturer's name, model or other code it must be interpreted to include other manufacturers' product

of like and equal quality whether the words “or equal” are included or not unless specifically stated otherwise.

4. Wherever a particular manufacturer’s product is required, to the exclusion of all others, appropriate language is included in the Contract Documents.
5. Wherever the terms, “as directed”, “ordered”, “permitted”, “designate”, “as approved”, “approved equal”, “or equal”, “acceptable” and other words of similar meaning which authorize an exercise of judgement are used in the Contract Documents, such judgment is vested only in the CO or designated representative.
6. When a particular manufacturer’s product is used, the Basis of Design and Section 01 25 00 SUBSTITUTIONS must be followed and be in conformance.

1.6 CORRESPONDENCE

- A. Contract correspondence must be directed to the CO with a copy to the COR.
- B. Send submittals directly to the COR with a copy of the transmittal letter to the CO.

1.7 LIST OF SUBCONTRACTORS

- A. The Contractor must, within 10 calendar days after award, furnish to the CO with a copy to the COR, a list of subcontractors showing the type of work each will perform. If all subcontracts have not been awarded when the initial list is submitted, the Contractor must update the list monthly.

1.8 WORK NOT INCLUDED

- A. Items noted on the drawings, details, or schedules as “Not in Contract” (“N.I.C.”) are not included in this contract.

1.9 WORK COVERED BY CONTRACT DOCUMENTS

- A. Project Identification: Addison Airport – Replace Fogging Cab Glass
- B. FAA's Lead Project Engineer: James E-CTR Arnold
- C. CO (Contracting Officer): Erin Crabtree
- D. The Work consists of the construction of the following:
 1. Traffic Control
 2. Cab Glass Replacement

1.10 WORK PHASES

- A. The Work must be conducted in multiple phases. The Contractor will be responsible for determining the sequence of operation to maintain security of the airport and the construction site.
- B. Before commencing Work, submit a schedule showing the sequence, commencement and completion dates, and move-in dates of FAA's personnel for all of the Work. See Section 01 32 00 Project Schedule.

- C. Construction Work stages shall be defined by the Contractor. The following sequence is only for the main construction site and does not consider work on the airport. It is provided as General Information Only.
 - 1. Traffic Control
 - 2. Crane Lifting
 - 3. Cab Glass Replacement Construction
 - 4. Remove Temporary Structures and Devices

1.11 USE OF PREMISES

- A. Contractor shall have full use of premises for construction operations, including use of Project site, during construction period. Contractor's use of premises is limited only by FAA's right to perform work or to retain other contractors on portions of Project.
- B. Use of Site: Limit use of premises to areas within the Contract limits indicated. Do not disturb portions of Project site beyond areas in which the Work is indicated.

1.12 WORK RESTRICTIONS

- A. The standard work schedule consists of 40 hours per week, Monday through Thursday, excluding U.S. Federal holidays. The Contractor must coordinate specific normal working hours with the COR. Contractor requests to work outside normal working hours require COR approval. However, the COR has full discretion to approve or disapprove or withdraw approval of requests. If the Contractor desires to work outside normal hours (including Fridays, Saturdays, Sundays, and holidays) submit written request to the COR at least 48 hours in advance. Some typical constraints on working outside normal working hours are:
 - 1. The Contractor's request must be made at least two days in advance (e.g., request received by close of business Wednesday for work on following Saturday). Prior to submitting the request, the Contractor must coordinate as needed (such as utility outages) and have all required people and materials for the work that will be performed.
 - 2. A Contractor with quality or safety problems (as determined by the COR) will be restricted to normal working hours. Contractors will not work outside of normal working hours if they are not present on the job site during normal working hours.
 - 3. A Contractor who fails to correct deficiencies within a reasonable time (as determined by the COR) will be restricted to normal working hours or may be allowed to work outside normal working hours only to correct those deficiencies.
 - 4. The Contractor must schedule his work to cause the least amount of interference to normal activities.
- B. Work at Cab: Work inside the cab and outside of cab glass must occur between 0800 to 1300 local time and be coordinated with the resident engineer. Cab glass removal and replacement activities must not occur when exterior temperature is above 80 degrees Fahrenheit or during bad weather.
- C. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by FAA or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:
 - 1. Notify FAA COR not less than 10 days in advance of proposed utility interruptions.

2. Do not proceed with utility interruptions without FAA and utility company's written permission.
- D. Nonsmoking Building: Smoking is not permitted inside the buildings or within 25 feet of the building entrances, operable windows or openings, outdoor air intakes throughout the duration of the construction.
- E. Limits on Operations: The FAA has established moratorium dates for construction activity at critical facilities including Lawton ATCT. The intent is to minimize the possibility of any activity that may have an adverse impact on the ability of FAA to perform its operational activities. Moratorium dates may change without notice. Work on new tower site is expected to continue but activities that could potentially impact the operation of the existing ATCT will not be permitted, including duct bank work. The anticipated moratorium dates are attached to this section.
 1. All construction activity during moratorium periods must be approved in advance by the FAA.
 - a. Submit items of work to be performed during moratorium dates no later than forty-five (45) days prior to the moratorium dates.
 - b. Activities that have, in the sole opinion of the FAA, potential to negatively impact FAA operations will not be approved.
 - c. A written waiver will be provided by FAA to the Contractor outlining the allowable work items. No additional time or cost will be allowed for such denial.
 2. Moratoriums, holidays and weekends are not included in determining the period of performance.

1.13 UTILITY CONTRACTOR LICENSING REQUIREMENTS

- A. Contractor must comply with all state and local requirements for construction of utilities.

1.14 SPECIFICATION FORMATS AND MISCELLANEOUS PROVISIONS

- A. Specification Format: The Specifications are organized into Divisions and Sections using the 50-division format and CSI/CSC's "MasterFormat" numbering system.
 1. Section Identification: The Specifications use Section numbers and titles to help cross-referencing in the Contract Documents. Sections in these Specifications are in numeric sequence; however, the sequence is incomplete because all available Section numbers are not used. Consult the table of contents at the beginning of the Project Manual to determine numbers and names of Sections in the Contract Documents.
 2. Division 01: Sections in Division 01 govern the execution of the Work of all Sections in the Specifications.
- B. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:
 1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be inferred as the sense requires. Singular words shall be interpreted as plural and plural words shall be interpreted as singular where applicable as the context of the Contract Documents indicates.

2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood are to be performed by Contractor. Occasionally, the indicative or subjunctive mood may be used in the Section Text for clarity to describe responsibilities that must be fulfilled indirectly by Contractor or by others when so noted.
 - a. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.

1.15 PERMITS AND FEES

- A. Contractor is responsible for applying for utility services, obtaining required permits, and payment for any associated fees. Compliance is required with the conditions of all permits that have been issued. All fees must be paid by the Contractor.
- B. Contractor is responsible for paying all utility service charges associated with the construction of the project until CAI or as otherwise agreed to by CO. This includes temporary and permanent utilities, permits, inspection fees, connection fees and equipment to be installed by utility companies. This allocation of financial responsibility applies to all utilities as well as City and County agencies and entities. Contractor is responsible for following up with the issuing authority after submittal.
- C. Listed below are the agencies that govern this project. Unless otherwise noted, Contractor is responsible for applying for utility services, obtaining required permits, and payment for any associated fees. Compliance is required with the conditions of all permits that have been issued. Additional permits may need to be obtained by the Contractor, and all fees must be paid by the Contractor.
 1. Sanitary Sewer - Permit approval will be provided to the Contractor. Contractor must comply with all provisions of present approval and the permit plans or obtain new approval. Contractor will pay for connection fees and establish customer service which will be transferred to FAA at no cost at completion of the construction contract.
 2. Water - Permit approval will be provided to the Contractor. Contractor must comply with all provisions of present approval and the permit plans or obtain new approval. Contractor will pay for connection fees and establish customer service which will be transferred to FAA at no cost at completion of construction contract.
 3. Erosion Control –Permit approval and conditions will be provided to the Contractor. It is the responsibility of the Contractor to notify local jurisdiction prior to beginning construction activities.
 4. Stormwater – Permit approval and conditions will be provided to the Contractor.
 5. Erosion control:
 - a. Permit approval and conditions will be provided to the Contractor. It is the responsibility of the Contractor to submit and obtain a new permit as operator of the site. All costs to submit the new permit will be the responsibility of the Contractor. The Contractor is responsible for maintaining and complying with the approved permit.
 - b. The contractor must follow the erosion control pans that are included on the Site drawings. During construction, the Contractor is responsible for following and implementing the approved erosion control plan. Any fees resulting from violations of the erosion control will be paid by the prime Contractor without any right of reimbursement from the Government.

6. Airspace– All permanent and temporary work, including cranes, must comply with FAA Obstruction Evaluation / Airport Airspace Analysis (OE/AAA) requirements. Contractor must file a Request for Construction for permanent structures on Airport. The Request should be submitted online. The website for more information and to obtain the appropriate forms is <https://oeaaa.faa.gov/>. Erection of the crane will not be permitted without approval of the form. No claim for delay will be allowed for failure to submit the form in a timely manner. Allow a minimum of sixty (60) days for processing. Contractor is responsible for following up with the issuing office after submittal.
7. Fuel Storage Tank and Piping – Contractor must apply and obtain permits and pay all fees.
8. Telephone and Electric Service – Obtain permits and pay all fees associated with temporary and permanent services which will be transferred to FAA at no cost at completion of construction contract.
9. Air Permit –The emergency generators will be subject to 40 CFR 60, Subpart IIII, and the facility will need to comply with all applicable emissions limits, fuel requirements, testing, monitoring, recordkeeping, and reporting requirements. Exemption letter and conditions will be provided to the Contractor. It is the responsibility of the Contractor to follow 40 CFR 60, Subpart III as well as comply with all applicable emissions limits, fuel requirements, testing, monitoring, recordkeeping, and reporting requirements related to any activities involving the emergency generators.
10. The contractor will be responsible for any additional fees, payments or applications required for the site utilities that have not been identified above.

D. Building Permit Applications

1. Contractor will not be required to obtain a Building Permit.

E. Traffic Control

1. Contractor is responsible for obtaining required permits, and payment for any associated fees. All fees must be paid by the Contractor.

F. Certificate Of Occupancy

1. Contractor will not be required to obtain a Certificate of Occupancy.

1.16 FAA'S OCCUPANCY REQUIREMENTS

- A. The FAA may occupy or use any completed or partially completed portion of the Work at any stage and, if the FAA chooses such partial occupancy, the Contractor and FAA must designate by an agreement the conditions of such partial occupancy. Such partial occupancy or use may commence whether or not the portion is substantially complete, provided the FAA and Contractor have accepted in writing the responsibilities assigned to each of them by the COR for payments, retainage if any, security, maintenance, heat, utilities, damage to the Work and insurance, and have agreed in writing concerning the period for correction of the Work and commencement of warranties required by the Contract Documents. Consent of the Contractor to partial occupancy or use by the FAA must not be unreasonably withheld.

1.17 UNCOVERING AND CORRECTION OF WORK

A. Uncovering Work

1. If any portion of the Work is covered contrary to the COR's request or to requirements specifically expressed in the Contract Documents, it must, if required in writing by the

COR be uncovered for his observation and be recovered (if corrections are not required) or be corrected, if applicable, at the Contractor's expense without change in the Contract Time.

B. Correction of Work

1. If, within two years after the date of Substantial Completion of the Work or designated portion thereof, or after the date for commencement of warranties established above, or by terms of an applicable special warranty required by the Contract Documents, any of the work is found to be not in accordance with the requirements of the Contract Documents, the Contractor must correct it promptly after receipt of written notice from the FAA to do so unless the FAA has previously given the Contractor a written acceptance of that specific condition.
 - a. This period of two years must be extended with respect to portions of Work first performed after Substantial Completion by the period of time between Substantial Completion and the actual performance of the Work.
 - b. This obligation survives acceptance of the Work under the Contract and termination of the Contract. The FAA must give such notice within a reasonable amount of time after discovery of the condition.
2. The Contractor must remove from the site portions of the Work that are not in accordance with the requirements of the Contract Documents and are neither corrected by the Contractor nor accepted by the FAA.
3. If the Contractor fails to correct nonconforming Work within a reasonable time, the FAA may correct it in accordance with General Provisions. If the Contractor does not proceed with correction of such nonconforming Work within a reasonable time fixed by written notice from the COR, the FAA may correct or remove such nonconforming work and all costs for such corrections or removals must be assessed against the Contractor.
4. The Contractor must bear the cost of correcting destroyed or damaged Work, whether completed or partially completed, of the FAA or separate contractors caused by the Contractor's performing correction or removal of Work which is not in accordance with the requirements of the Contract Documents.
5. Nothing contained herein shall be construed to establish a period of limitation with respect to other obligations that the Contractor might have under the Contract Documents.
 - a. Establishment of the time period of two years as described above relates only to the specific obligation of the Contractor to correct the Work.
 - b. This requirement has no relationship to the time within which the obligation to comply with the Contract Documents may be sought to be enforced, nor to the time within which proceedings may be commenced to establish the Contractor's liability and damages with respect to the Contractor's obligations other than specifically to correct the Work.

1.18 INSURANCE LIMITS

- A. In addition to AMS Paragraph 3.4.1-10, Insurance - Work on a Government Installation, during the term of the Contract, the Contractor and each subcontractor must at their own expense purchase and maintain the following minimum insurance requirements in companies properly licensed and satisfactory to the Contracting Officer:
1. Automobile and Truck Liability
 - a. \$200,000 - bodily injury per person, not to exceed \$500,00 per occurrence.
 - b. \$100,000 - property damage per occurrence.

2. Comprehensive General Liability - \$500,000 - Combined bodily injury and property damage per occurrence.
 3. Workmen's Compensation - \$100,000 or statutory, whichever is greater.
 4. Umbrella or Excess Liability - \$1,000,000 combined single limits bodily injury and property damage.
- B. The policy must name "The United States of America, acting by and through the Federal Aviation Administration as an additional insured with respect to operations performed under this contract.
- C. Insurance certificates must contain a provision that coverage afforded under the policies will not be canceled until at least 30 days prior written notice has been given to the Contracting Officer.
- D. It is agreed that the Federal Aviation Administration, the property owner(s) and their representatives will be held harmless by the Contractor for any loss or damage to sheds, tools, equipment, property and materials of the Contractor, and his subcontractors, their servants and employees, it being understood that the Contractor may at his expense carry any insurance which may be required to provide the necessary protection against such loss or damage.
- E. The Contractor must not commence work under the contract until all the insurance required hereunder is obtained and such insurance has been approved by the Contracting Officer. Approval of the insurance by the Contracting Officer shall not relieve or decrease the liability of the Contractor.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 10 00

SECTION 01 10 15

ACRONYMS AND DEFINITIONS

PART 1 - GENERAL

1.1 ACRONYMS AND DEFINITIONS

- A. When in the Special Provisions, this specification, drawings, specifications, or documents pertaining to this contract, the following terms are used; the intent and meaning shall be as specified herein.
1. AMS - Acquisition Management System
 2. AMSL - Above Mean Sea Level
 3. AOA - Air Operations Area
 4. ARCHITECT - Architectural Engineering Firm of Record
 5. ATCT - Airport Traffic Control Tower
 6. CAI - Contractors Acceptance Inspection or Substantial Completion
 7. CFM - Contractor-Furnished Material
 8. CO - FAA Contracting Officer
 9. Contr. - Contractor
 10. COR - FAA Contracting Officer Representative (Fulltime Onsite Representative of the FAA and is also referred to as the (COTR) Contracting Officer's Technical Representative
 11. ENGINEER - Engineering Firm of Record
 12. FAA - Federal Aviation Administration (Owner)
 13. GFE - Government-Furnished Equipment
 14. GFM - Government-Furnished Material
 15. GOVERNMENT – FAA
 16. IAW - In Accordance With
 17. LAW – Lawton Fort Sill Regional Airport
 18. MSL - Mean Sea Level
 19. NIC – Not in Contract
 20. NEC - National Electric Code
 21. NTP - Notice to Proceed
 22. OSHA - Occupational Safety and Health Administration
 23. Owner - FAA
 24. RE - Resident Engineer.
 25. RWY - Runway
 26. Sponsor - Airport Owner or Airport Authority
 27. TWY - Taxiway
 28. U/L or UL - Underwriters Laboratories

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 10 15

SECTION 01 10 20

CONSTRUCTION ADMINISTRATION FORMS

PART 1 - GENERAL

- A. **INDEX OF CONSTRUCTION ADMINISTRATION FORMS:** The following construction forms are required for this Project and are available from FAA.

Alteration RFI Standard Form

Shop Drawing, Product Data, or Sample Transmittal

Certificate of Substantial Completion (CoSC)

Substantial Completion Acceptance (SCA)

Partial Occupancy/Use Agreement (POUA)

Job Memorandum (JM)

Hot Work Permit

Lock Out/Tag Out Procedure (See Division 26)

Contractor must submit a copy of Airspace Form 7460.1 to COR.

Contractor must submit Airspace Study to oeaaa.faa.gov and notify COR.

Contractor must submit latest version of the NFPA System Record of Inspection and Testing Form and NFPA Certificate of Completion form as required by NFPA 72.

Contractor shall be responsible to follow up with airport division within 2 weeks of submission to verify receipt and ensure timely processing of the form.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

- A. During the administration of the Contract, the Contractor will be required to complete various construction administration forms as a part of the Management System. These forms are identified above and will be issued at the Pre-Construction Conference. These forms may be revised during the construction period and the Contractor will be required to comply with any such revisions.

END OF SECTION 01 10 20

SECTION 01 25 00

SUBSTITUTION PROCEDURES

PART 1 - GENERAL

1.1 DEFINITIONS

- A. Substitution: Any product or material that is submitted that is not the exact make and model number of the design basis shall be considered a substitution. This includes products that are from the same manufacturer, but are different models. All substitutions shall follow the substitution procedures listed herein.
- B. Known Acceptable Source: A manufacturer of a particular product or material that has been utilized successfully on past FAA projects. This is not an indication that a particular manufacturer will meet the requirements of each FAA project, only that they have been found to meet the requirements on past projects.
- C. When a material, article, or equipment is designated by brand name, and more than one brand name is listed, it will be understood that the design is based on one of the brand names listed.
- D. Where an item is identified by a particular manufacturer's name, model, or other code, it must be interpreted to include other manufacturers' product of like and equal quality whether the words "or approved equal" are included or not unless specifically stated otherwise.

1.2 ACTION SUBMITTALS

- A. Submit substitution proposals on attached Product Substitution Request Form.

1.3 SUBSTITUTIONS

- A. Submission of request for substitution shall constitute a representation by the Contractor that he:
 - 1. Has investigated the proposed product and determined that it is equal to or better than the specified product. Absence of an explicit comparison of any characteristic of the proposed product to the specified product shall constitute a representation that the proposed product is equal to or better than the specified product with regard to that characteristic.
 - 2. Will provide the same warranty for the proposed product as for the specified product.
 - 3. Will coordinate the installation and make other changes which may be required for the work to be complete in all respects, including:
 - a. Redesign.
 - b. Additional components and capacity required by other work affected by the change.
 - c. Update BIM.
 - 4. Waives all claims for additional costs and time extensions which subsequently may become apparent and which are caused by the change.
 - 5. Will reimburse the Government for additional costs for evaluation of the substitution request, redesign if required, and reapproval by authorities having jurisdiction if required.

- B. Substitutions will not be considered when acceptance would require substantial revision of the contract documents.
- C. Substitutions will not be considered when they are indicated or implied on shop drawing or product data submittals without separate written request.
- D. Substitution requests will not be considered when submitted directly by subcontractor or supplier.
- E. Substitution Request Procedure: Submit written request with complete data substantiating compliance of the proposed product with the requirements of the contract documents.
 - 1. Submit request to the Contracting Officer Representative (COR).
 - 2. Digitally submit 3 copies of each request and accompanying data.
 - 3. Submit all requests on a standard form provided.
 - 4. Only one request for substitution will be considered for each product.
- F. Data Required with Substitution Request: Provide at least the following data:
 - 1. Identify product by specification section and paragraph number.
 - 2. Manufacturer's name and address, trade name and model number of product (if applicable), and name of fabricator or supplier (if applicable).
 - 3. Complete product data.
 - 4. A list of other projects on which the proposed product has been used, with project name, and the design professional's name.
 - 5. An itemized comparison of the proposed product to the specified product.
 - 6. Net amount of change to the contract sum.
 - 7. List of maintenance services and replacement materials available.
 - 8. Statement of the effect of the substitution on the construction schedule.
 - 9. Description of changes that will be required in other work or products if the substitute product is approved.
- G. The COR will determine acceptability of the proposed substitution.
- H. When the proposed substitution is not accepted, provide the product (or one of the products, as the case may be) specified.

PART 2 - PRODUCTS

2.1 NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 25 00

PRODUCT SUBSTITUTION REQUEST FORM

Note: This form to be used by General Contractor only. Requests by others will be returned with no response.

Project:

Location:

Government:

Date:

We hereby submit for your consideration the following substitution instead of the item specified or shown on the Drawings:

Section Number:	Paragraph	Specified item
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Proposed Substitution:

Attach complete product data, drawings and descriptions of products, with fabrication and installation details. Provide laboratory tests if applicable.

Provide sample, if applicable. Indicate if sample will be provided under separate cover.

Include complete information on changes to Drawings and/or Specifications that proposed substitution will require for its proper installation.

Fill in blanks below: *(Include attachments if space is insufficient. Failure to provide information will void submittal)*

A. Reason(s) for proposed substitution: *(check all that apply)*

- ☐ 1. Request is equivalent to product/material/assembly specified. *(Note: Attach technical documentation)*
- ☐ 2. Specified product or method cannot be provided within the Contract time. *(Note: This request will not be considered if the product or method cannot be provided as a result of the Contractor's failure to pursue the Work promptly, or to coordinate the various activities properly, or if the Contractor fails to place timely orders)*
- ☐ 3. Specified product or method cannot receive necessary approval by authority having jurisdiction, and Contractor certifies that the requested substitution can be approved. *(Note: Attach approval documentation)*
- ☐ 4. A substantial advantage is offered the Government, in terms of cost, time, energy conservation or other considerations of merit, after deducting redesign and evaluation costs of other work by the Government or separate contractors and similar considerations.

- _____ 5. Specified product or method cannot be provided in a manner which is compatible with other materials of the Work, and the Contractor certifies that the substitution will overcome the incompatibility.
- _____ 6. Specified product cannot be properly coordinated with other materials in the Work, and the Contractor certifies that the proposed substitution can be properly coordinated.
- _____ 7. Specified product or method cannot receive a warranty as required by Contract Documents, and Contractor certifies that the proposed substitution can receive required warranty.

B. Does the substitution affect dimensions or details shown on Drawings:

_____ No

_____ Yes (*Note: Attach marked up prints of drawings showing changes required*)

C. What effect does the substitution have on other trades?

D. Compare significant qualities of proposed substitution with those of work or product originally specified or shown on drawings. Include elements such as size, weight, durability, performance, visual effect, etc.

E. Coordinate information. Include all changes required in other elements of the work in order to accommodate the substitution, including work performed by Government or separate contractors.

F. State effect substitution will have on the work schedule in comparison to the schedule which would prevail without the proposed substitution. State the effect of the proposed substitution on Contract Time.

G. Provide complete cost information, including a proposal of any net change in the Contract Amount.

H. Manufacturer's warranties of the proposed substitution and specified items are:

_____ Same

_____ Different (*Note: Explain on attachment*)

The undersigned Contractor certifies its opinion that, after thorough evaluation, the proposed substitution will result in work that in every significant respect will

be equivalent to or superior to the work required by the original Contract Documents and that it will perform adequately in the application indicated.

Rights to additional payment or time because of failure of the substitution to perform adequately are hereby waived.

The undersigned hereby agrees to pay in full for any changes to design, including detailing and engineering costs caused by the requested substitution.

Submitted by: *(Note: Submittal void and will be discarded if unsigned or if signed by entity other than Contractor)*

Signature: _____ (Contractor's authorized representative)

(Title)

Firm Name: _____

Date: _____

For use by Contracting Officer Representative:

____ Accepted

____ Accepted as Noted

____ Not Accepted

____ Received too late

By: _____ Date: _____

(Contracting Officer Representative)

By: _____ Date: _____

(Contracting Officer Representative)

Remarks:

SECTION 01 31 00

PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

1. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following: General project coordination procedures.
2. Administrative and supervisory personnel.
3. Requests for Information (RFIs).

1.2 DEFINITIONS

- A. RFI: Request for Information from COR, Designer, or Contractor seeking information from each other during construction.

1.3 ACTION SUBMITTALS

- A. Submit RFI requests as specified herein.

1.4 INFORMATIONAL SUBMITTALS

- A. Key Personnel: Within 14 calendar days of Notice to Proceed, submit a list of key personnel assignments, including superintendent and other personnel in attendance at Project site. Identify individuals by name and describe their duties and responsibilities; list addresses and telephone numbers, including office, and cellular telephone numbers and email addresses. Provide names, addresses, and telephone numbers of individuals assigned as standbys in the absence of individuals assigned to Project.
 1. Post copies of list in project meeting room, in temporary field office, and by each temporary telephone. Keep list current at all times.

1.5 CONTRACTOR PERSONNEL REQUIREMENTS

- A. Project Manager with a minimum education level of a Bachelor of Science (BS) degree in engineering (from an accredited institution), or a technical degree, and five (5) years of experience with coordinating subcontractors on projects with complex mechanical, electrical, and control systems in the heavy construction industry. The project manager must provide a project management certification from a recognized institute or organization, such as, Project Management Institute. The project manager and the project superintendent cannot be the same individual.
- B. Project Superintendent with a minimum of five (5) years of experience in coordinating mechanical, electrical, and control subcontractors in heavy construction industry. The superintendent must maintain a physical presence at the project site throughout the construction project unless approved by the FAA. The project superintendent and the project manager cannot be the same individual.

1.6 COORDINATION

- A. Coordination: Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections that depend on each other for proper installation, connection, and operation.
 - 1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
 - 2. Coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair.
 - 3. Make adequate provisions to accommodate items scheduled for later installation.
- B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
 - 1. Prepare similar memoranda for COR and separate contractors if coordination of their Work is required.
- C. Coordination Meetings: Host regular weekly (or more frequent) progress meetings in accordance with Section 01 31 19 PROJECT MEETINGS. Attendance is mandatory by all Team members to maintain the coordination and construction schedules.
- D. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:
 - 1. Preparation of Contractor's construction schedule in accordance with Section 01 32 00 Project Schedule and preparation of the schedule of values.
 - 2. Installation and removal of temporary facilities and controls in accordance with Section 01 50 00 Temporary Facilities and Controls.
 - 3. Delivery and processing of submittals in accordance with Section 01 33 00 Submittal Procedures.
 - 4. Progress meetings and pre-installation conferences in accordance with Section 01 32 29 Project Meetings.
 - 5. Project closeout activities in accordance with Section 01 77 00 PROJECT CLOSEOUT
- E. Conservation: Coordinate construction activities to ensure that operations are carried out with consideration given to conservation of energy, water, and materials. Coordinate use of temporary utilities to minimize waste.
 - 1. Salvage materials and equipment involved in performance of, but not actually incorporated into, the Work. Refer to other Sections for disposition of salvaged materials that are designated as FAA's property.

1.7 REQUESTS FOR INFORMATION (RFIS)

- A. Immediately on discovery of the need for additional information or interpretation of the Contract Documents, Contractor shall prepare and submit electronically an RFI in the form specified to the COR.
 - 1. COR will return RFIs submitted by other entities controlled by Contractor with no response.

2. Coordinate and submit RFIs in a prompt manner so as to avoid delays in Contractor's work or work of subcontractors.
 3. Limit each RFI to one subject.
- B. Content of the RFI: Include a detailed, legible description of item needing information or interpretation and the following:
1. Project name.
 2. Project number.
 3. Date.
 4. Name of Contractor.
 5. Name of COR.
 6. RFI number, numbered sequentially.
 7. RFI subject.
 8. Specification Section number and title and related paragraphs, as appropriate.
 9. Drawing number and detail references, as appropriate.
 10. Field dimensions and conditions, as appropriate.
 11. Contractor's suggested resolution. If Contractor's solution(s) impacts the Contract Time or the Contract Sum, Contractor shall state impact in the RFI.
 12. Contractor's signature.
 13. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
 - a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.
- C. RFI Forms: Refer to Section 01 10 20 CONSTRUCTION ADMINISTRATION FORMS.
- D. COR's Action: COR will review each RFI, determine action required, and respond. Allow fourteen (14) calendar days for COR's response for each RFI. RFIs received by COR after 1:00P.M. local time will be considered as received the following working day.
1. The following RFIs will be returned without action:
 - a. Requests for approval of submittals.
 - b. Requests for approval of substitutions.
 - c. Requests for coordination information already indicated in the Contract Documents.
 - d. Requests for adjustments in the Contract Time or the Contract Sum.
 - e. Requests for interpretation of Designer's actions on submittals.
 - f. Incomplete RFIs or inaccurately prepared RFIs.
 2. COR's action may include a request for additional information, in which case COR's time for response will date from time of receipt of additional information.
 3. COR's action on RFIs that may result in a change to the Contract Time or the Contract Sum may be eligible for Contractor to submit Change Proposal according to Division 01 Section "Contract Modification Procedures."
 - a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify COR in writing within four (4) calendar days of receipt of the RFI response.
- E. On receipt of COR's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify COR within seven (7) calendar days if Contractor disagrees with response.

- F. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log weekly. Include the following:
1. Project name.
 2. Name and address of Contractor.
 3. RFI number including RFIs that were dropped and not submitted.
 4. RFI description.
 5. Date the RFI was submitted.
 6. Date COR's response was received.
 7. Identification of related Minor Change in the Work, Construction Change Directive, and Proposal Request, as appropriate.
 8. Identification of related Field Order, Work Change Directive, and Proposal Request, as appropriate.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 01 31 00

SECTION 01 31 19

PROJECT MEETINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Description of Work: Schedule and hold pre-construction, progress and pre-installation meetings throughout construction of Work.
- B. Other meetings are required, including:
 - 1. Project Schedule update meetings: See Section 01 32 00 PROJECT SCHEDULE.
 - 2. Quality control and coordination meetings: See Section 01 40 00 CONTRACTOR QUALITY CONTROL.
 - 3. Preconstruction Safety meeting: See Section 01 35 29 SAFETY AND HEALTH;

1.2 INFORMATION SUBMITTALS

- A. Submit meeting schedules, agendas and minutes as specified herein.

1.3 PROJECT MEETINGS

- A. General: Schedule and conduct meetings and conferences at Project site.
 - 1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date and time of each meeting. Notify COR of scheduled meeting dates, times and agenda at least five days in advance of meeting.
 - 2. Agenda: Prepare the meeting agenda. Distribute the agenda to all invited attendees after COR approves agenda and at least 24 hours before the meeting.
 - 3. Minutes: Contractor shall record significant discussions and agreements achieved. Distribute the meeting minutes to everyone concerned, including COR, within three days of the meeting.
- B. Preconstruction Conference: COR will schedule and conduct a preconstruction conference before starting construction, at a time convenient to the government and the contractor, but no later than 14 days prior to NTP.
 - 1. Conduct the conference to review responsibilities and personnel assignments.
 - 2. Attendees: Authorized representatives of COR's; project manager, project superintendent; major subcontractors; suppliers; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 3. Agenda: Discuss items of significance that could affect progress, including the following:
 - a. Tentative construction schedule.
 - b. Phasing/sequencing.
 - c. Permits
 - d. Critical work sequencing and long-lead items.
 - e. Designation of key personnel and their duties.
 - f. Lines of communications.

- g. Procedures for processing field decisions and Change Orders.
 - h. Contract requirements by CO.
 - i. Procedures for RFIs.
 - j. Procedures for testing and inspecting.
 - k. Procedures for processing Applications for Payment.
 - l. Distribution of the Contract Documents.
 - m. Submittal procedures.
 - n. Sustainable design requirements.
 - o. Preparation of record documents.
 - p. Use of the premises.
 - q. Work restrictions.
 - r. Working hours.
 - s. FAA's occupancy requirements.
 - t. Responsibility for temporary facilities and controls.
 - u. Procedures for moisture and mold control.
 - v. Procedures for disruptions and shutdowns.
 - w. Construction waste management and recycling.
 - x. Parking availability.
 - y. Office, work, and storage areas.
 - z. Equipment deliveries and priorities.
 - aa. First aid.
 - bb. Security.
 - cc. Progress cleaning.
 - dd. Environmental requirements and procedures, including but not limited to:
 - 1) Erosion and Sediment control;
 - 2) Solid Waste Management Plan;
 - 3) IAQ Management Plan;
 - 4) Procedures for noise and acoustics management;
 - 5) Environmental Management Plan;
 - 6) Commissioning.
4. Minutes: Contractor shall record and distribute meeting minutes.
- C. Pre-installation Conferences: Conduct a preinstallation conference at Project site before each construction activity that requires coordination with other construction.
- 1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise COR of scheduled meeting dates.
 - 2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:
 - a. Contract Documents.
 - b. Options.
 - c. Related RFIs.
 - d. Related Change Orders.
 - e. Purchases.
 - f. Deliveries.
 - g. Submittals.
 - h. Review of mockups.
 - i. Possible conflicts.

- j. Compatibility problems.
 - k. Time schedules.
 - l. Weather limitations.
 - m. Manufacturer's written recommendations.
 - n. Warranty requirements.
 - o. Compatibility of materials.
 - p. Acceptability of substrates.
 - q. Temporary facilities and controls.
 - r. Construction waste management and recycling.
 - s. Indoor air quality management.
 - t. Space and access limitations.
 - u. Regulations of authorities having jurisdiction.
 - v. Testing and inspecting requirements.
 - w. Installation procedures.
 - x. Coordination with other work.
 - y. Required performance results.
 - z. Protection of adjacent work.
 - aa. Protection of construction and personnel.
 - 3. Contractor shall record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
 - 4. Reporting: Distribute minutes of the meeting to each party present and to other parties requiring information.
 - 5. Unless otherwise directed by the Contracting Officer, do not proceed with installation if the conference is not successfully concluded.
 - 6. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.
- D. Progress Meetings: Conduct progress meetings at weekly intervals, or at such other time and frequency as are acceptable to COR. Coordinate dates of meetings with preparation of payment requests.
- 1. Attendees:
 - a. Representatives of COR, and their consultants as applicable; Contractor and its superintendent; each subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities.
 - b. All participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
 - a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Contractor's Construction Schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
 - 1) Review schedule for next period.
 - b. Schedule Updating: Revise construction schedule after each meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with report of each meeting.

- c. Review present and future needs of each entity present, including the following:
 - 1) ORM, Hazards, and Risk.
 - 2) Interface requirements.
 - 3) Access.
 - 4) Site utilization.
 - 5) Sequence of operations.
 - 6) Temporary facilities and controls.
 - 7) Weather delays.
 - 8) Quality and work standards.
 - 9) Status of submittals.
 - 10) RFIs.
 - 11) Deliveries.
 - 12) Off-site fabrication.
 - 13) Pending changes.
 - 14) Status of Change Orders.
 - 15) Pending claims and disputes.
 - 16) Status of correction of deficient items.
 - 17) Status of proposal requests.
 - 18) Field observations.
 - 19) Work hours.
 - 20) Documentation of information for payment requests.
 - 21) Progress cleaning.
 - 3. Minutes: Contractor shall record the meeting minutes.
 - 4. Reporting: Distribute minutes of the meeting to each party present and to parties who should have been present within 2 days.
- E. Project Closeout Conference: Schedule and conduct a Project closeout conference, at a time convenient to the COR, but no later than 60 days prior to the scheduled date of Substantial Completion.
 - 1. Conduct the conference to review requirements and responsibilities related to Project closeout.
 - 2. Attendees: Authorized representatives of COR; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the meeting. Participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 3. Agenda: Discuss items of significance that could affect or delay Project closeout, including the following:
 - a. Preparation of record documents.
 - b. Procedures required prior to inspection for Substantial Completion and for final inspection for acceptance.
 - c. Submittal of written warranties.
 - d. Requirements for preparing sustainable design documentation.
 - e. Requirements for preparing operations and maintenance data.
 - f. Requirements for demonstration and training.
 - g. Preparation of Contractor's punch list.
 - h. Procedures for processing Applications for Payment at Substantial Completion and for final payment.
 - i. Submittal procedures.
 - j. Coordination of separate contracts.

- k. FAA's partial occupancy requirements.
 - l. Installation of FAA's furniture, fixtures, and equipment.
 - m. Responsibility for removing temporary facilities and controls.
4. Minutes: Contractor shall record and distribute meeting minutes.

PART 2 - PRODUCTS

Not Used

PART 3 - EXECUTION

Not Used

END OF SECTION 01 31 19

SECTION 01 32 00

PROJECT SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
 - 1. Preconstruction project schedule.
 - 2. Construction project schedule.
 - 3. Periodic schedule updates
 - 4. Weekly progress meetings
 - 5. Time extensions
 - 6. Schedule of Values

1.2 DEFINITIONS

- A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They must start and finish on the planned early start and finish times.
 - 2. Predecessor Activity: An activity that precedes another activity in the network.
 - 3. Successor Activity: An activity that follows another activity in the network.
- B. Cost Loading: The allocation of the Schedule of Values for the completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by COTR.
- C. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- D. Critical Path: The longest connected chain of interdependent activities through the network schedule that establishes the minimum overall Project duration and contains no float.
- E. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail
- F. Float: The measure of leeway in starting and completing an activity
 - 1. Float time is not for the exclusive use or benefit of either FAA or Contractor, but is a jointly owned, expiring Project resource available to both parties as needed to meet schedule milestones and Contract completion date
 - 2. Activity float is the number of work days that an activity can be delayed without causing a delay to the "End Project" finish milestone.

3. Project float is the number of work days between the projected early finish and the Contract completion date milestone

G. Milestone: A key or critical point in time for reference or measurement

1.3 SUBMITTALS

- A. Qualification Data: For scheduling consultant or in-house scheduler
- B. Preconstruction Project Schedule: See section 3.1 for requirements
- C. Baseline Project Schedule: See section 3.2 for requirements
- D. Periodic Schedule Updates: See section 3.3 for requirements
- E. Weekly Meetings: See section 3.4 for requirements
- F. Schedule of Values: See section 3.6 for requirements

1.4 QUALITY ASSURANCE

- A. Scheduling Consultant Qualifications: Authorize a representative to prepare the schedule and update and produce reports. The authorized representative shall have experience scheduling construction projects similar in size and nature to this project. Representative shall have a comprehensive knowledge of CPM scheduling principles and application, with capability of producing CPM reports and diagrams within 24 hours of COR's request.
- B. In-House Option: FAA may waive requirement to retain a consultant if Contractor employs skilled personnel with experience in CPM scheduling and reporting techniques. Submit qualifications

PART 2 - PRODUCT

NOT USED

PART 3 - EXECUTION

3.1 PRECONSTRUCTION SCHEDULE REQUIREMENTS

- A. General
 1. Prepare a detailed Preconstruction Project Schedule for the duration of the project and include all milestone activities
 2. Submit the Preconstruction Project Schedule to the Contracting Officer (CO) within 14 calendar days after contract award.
 3. The Preconstruction Project Schedule shall consist of a diagram or a bar chart showing the start and the finish dates of construction, as well as the major items to be constructed, what work is occurring, length of time anticipated for the activity and the flow of construction

- a. Diagram(s) shall show the order and interdependence of activities and the sequence in which the diagram will be followed to show how the start of a given activity is dependent on the completion of preceding activities and its completion restricts the start of following activities.
 - b. Diagram activities shall include, in addition to construction activities, the submittal, review and approval of samples of materials and shop drawings, the procurement of critical materials and equipment, fabrication of special materials and equipment and their installation and testing. All activities of the Government and others that affect progress, and contract required dates for completion of all parts of the work shall also be shown.
4. The Preconstruction Project Schedule shall consist of a minimum of 10 activities. The selection of activities shall be subject to the Contracting officer's approval. Scheduling software may be used to produce this schedule

3.2 CONSTRUCTION PROJECT SCHEDULE REQUIREMENTS

A. General

1. Prepare for approval a Project Schedule, as specified herein, pursuant to AMS Clause 3.2.2.3-56, SCHEDULES FOR CONSTRUCTION CONTRACTS.
2. Must be generated using Critical Path Method (CPM)
3. Provide in pdf format
4. The schedule must be submitted and approved prior to NTP.
5. Schedule must contain all contractual work and be broken down into sufficient detail to allow all interested parties insight into the project plan. Treat each floor or separate area as a separate numbered activity for each main element of the work.
6. Contractor management personnel shall actively participate in its development. Subcontractors and suppliers' designers, subcontractors, and suppliers working on the project shall also contribute in developing and maintaining an accurate project schedule.
7. Provide the schedule data to the COR. This data is considered to be additional supporting data in a form and detail required by the Contracting Officer pursuant to AMS 3.3.1-2 "Payments under Fixed-Price Construction Contracts". The receipt of a proper payment requests pursuant to AMS 3.3.1-19 "Prompt Payment for Construction Contracts" is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic files of the application for progress payment.
8. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail.

B. Activity Duration

1. Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. No activities shall have Original Durations (OD) greater than 20 work days or 30 calendar days.
2. Activity Original Durations shall be reasonable to perform the work item. Duration changes are prohibited unless justification is provided to and approved by the Contracting Officer.
3. When resubmittals are required, a new activity must be added to the schedule which reflects a new review time
4. Use "one workday" as the unit of time for individual activities. Indicate nonworking days and holidays incorporated into the schedule to coordinate with the Contract Time
5. For activities affected by weather, assign it to the calendar showing non-work days on a monthly basis, using the anticipated non – work days that shall be calculated by the

Contractor based up on the procedures defined in USA CE ER 415 -1 -15, Appendix A & B. The non-work days shall be selected at random across the weeks of the calendar. The assignment of the non-work days should be over a 7-day week since weather records are compiled on 7-day weeks, which will cause some of the weather related non-work days to fall on weekends.

C. Mandatory Activities

1. Include the following activities in the initial project schedule and all updates.
 - a. Submission, review and acceptance of all submittals and be broken down into four (4) categories
 - 1) Date of Submittal
 - 2) Submittal Review
 - 3) Procurement
 - 4) Delivery
 - b. Traffic Control Permitting
 - c. Mobilization
 - d. Performance of Cab Glass Work
 - e. Substantial Completion.
 - f. Contractor's pre-final inspection.
 - g. Correction of punch list from Contractor's pre-final inspection.
 - h. Government's inspection for Substantial Completion (Contractor Acceptance Inspection).
 - i. Correction of punch list from Government's inspection for Substantial Completion (Contractor Acceptance Inspection).
 - j. Submission and approval of O & M manuals.
 - k. Submission and approval of as-built drawings.
 - l. Final Cleaning.
 - m. Demobilization

D. Contract Milestones and Constraints

1. Milestone activities shall be used for significant project events including, but not limited to, project phasing, project start and end activities, and interim completion dates.
2. Mandatory constraints that ignore or affect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case-by-case basis.
3. Include, as a minimum the following milestones in the schedule.
 - a. Notice of Award
 - b. NTP
 - c. 100% completion of Traffic Control Plans
 - d. 100% completion of Cab Glass Building
 - e. CAI Complete
 - f. CAI Punch List Complete
 - g. End Project
4. Project Start Date Milestone and Constraint
 - a. The first activity in the project schedule shall be a start milestone titled "NTP," which shall have a "Start On" constraint date equal to the agreed upon NTP date between the FAA and contractor.
5. End Project Finish Milestone and Constraint
 - a. The last activity in the schedule shall be a finish milestone titled "End Project".

- b. The project schedule shall be constrained to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path. The Government is under no obligation to accelerate Government activities to support a Contractor's early completion.
- E. Open Ended Logic
 - 1. Only two (2) open ended activities are allowed: the first activity "NTP" shall have no predecessor logic, and the last activity "End Project" shall have no successor logic.
 - 2. Predecessor open-ended logic may be allowed in a time impact analysis upon the Contracting Officer's approval.
- F. Added and Deleted Activities
 - 1. Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer.
- G. Leads, Lags, and Start to Finish Relationships
 - 1. Lags shall be reasonable as determined by the Government and not used in place of realistic original durations, shall not be in place to artificially absorb float, or to replace proper schedule logic. Leads (negative lags) and Start to Finish (SF) relationships are prohibited.
- H. Percent Complete
 - 1. Update the percent complete for each activity started, based on the realistic assessment of physical completion. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management.
- I. Early Completion Schedule and the Right to Finish Early
 - 1. An Early Completion Schedule is a Baseline Project Schedule that indicates the scope of the required contract work will be completed before the contractually required completion date.
 - a. No baseline schedule indicating an Early Completion will be accepted. The time, in calendar days, between the schedule 'NTP' start milestone and its 'End Project' finish milestone shall match the contract's Period of Performance. The 'End Project' finish date, as a consequence, shall be the same as the contractually required completion date.
 - b. The Government is under no obligation to accelerate its own work items if early completion occurs, nor is it responsible to modify incremental funding (if applicable) for the project to meet the Contractor's accelerated work.
- J. Failure to achieve progress
 - 1. If the progress falls behind the approved project schedule for reasons other than those that are excusable within the terms of the Contract, the Contracting Officer may require submittal of a written recovery plan for approval. The plan shall detail how progress shall be recovered, including which activities will be accelerated by adding additional crews, longer work hours, extra work days, etc.
 - a. Artificially Improving Progress

- 1) Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Additional resources, manpower, and daily and weekly work hour changes proposed shall be evident at the work site and documented in the daily report along with the Schedule Narrative Report.
 - b. Failure to Perform
 - 1) Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and/or may result in corrective action directed by the Contracting Officer pursuant to AMS 3.2.2.3-56 "Schedules for Construction Contracts", AMS 3.10.6-6 "Default (Fixed-Price Construction)", and other Contract provisions
- K. Basis for payment and cost loading
 1. The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment.
 - a. Activity Cost Loading;
 - 1) Activity cost loading shall be reasonable and without front-end loading. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.
 - 2) Cost loading should equal the total contract value
 2. Withholdings / Payment Rejection
 - a. Failure to meet the requirements of this Section may result in the disapproval of the schedules or updates and subsequent rejection of payment requests until requirements are met.
 3. If the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount for each payment period until such revisions to the project schedule have been made

3.3 PERIODIC SCHEDULE UPDATE

- A. Periodic Schedule Updates
 1. Update the Project Schedule on a regular basis. An updated schedule is required with each pay application, and more frequently if requested by the Contracting Officer. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in Paragraph PERIODIC SCHEDULE UPDATE MEETINGS, below. These updates will enable the Government to assess Contractor's progress
 - a. Update information, including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete, is subject to the approval of the Government at the meeting
 - b. AS and AF data shall match the dates reported on the Contractor's Daily Construction Reports for an activity start or finish
- B. Periodic Schedule Update Meetings

1. Conduct periodic schedule update meetings, to review the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly. The Contractor's authorized scheduler shall organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. Provide a draft of the proposed narrative report and schedule PDF to the Government at least 2 workdays in advance of the meeting. The Contractor's Project Manager and scheduler shall attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen, and major subcontractors shall attend the meeting as required to discuss the project schedule and work. Following the periodic schedule update meeting, make corrections to the draft submission. Include only those changes approved by the Government in the submission and invoice for payment.
2. Update Submission Following Progress Meeting
 - a. Submit the complete Periodic Schedule Update, containing the approved progress, revisions, and adjustments, not later than 4 work days after the periodic schedule update meeting.
 - b. Submit final periodic update schedule in the original software format and pdf snapshot
 - c. Submission of Contractor Periodic Schedule Update shall be required for periodic progress payments

3.4 WEEKLY PROGRESS MEETINGS

- A. Conduct a weekly meeting with the Government, separate from the Periodic Schedule Update Meetings, for the purpose of jointly reviewing the actual progress of the project as compared to the planned progress and to review planned activities for the upcoming 3 weeks. Provide a 3-week look ahead schedule prior to the weekly meeting. Submit the 3-week look ahead a minimum of 2 days before the weekly meeting. At the weekly progress meeting, address the status of RFIs, RFPs, and Submittals.
- B. Contractor must take the meeting minutes of these meetings and distribute the minutes to attendees for comments. Submit minutes not later than 2 work days after the weekly meeting

3.5 TIME EXTENSIONS

- A. Provide a justification of delay to the Contracting Officer, in accordance with the Contract provisions and clauses, for approval within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP) to justify time extensions.
 1. Justification of Delay
 - a. Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify the schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date or interim completion dates. Multiple impacts shall be evaluated chronologically; each with its own justification of delay. With multiple impacts, consider concurrency of delay. A time extension and the schedule fragmet becomes part of the project schedule and future schedule updates upon approval by the Contracting Officer.
 2. Time Impact Analysis (Prospective Analysis)

- a. Prepare a time impact analysis for Government approval based on industry standard AACE 52R-06. Use a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If the Government determines the timeframe between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes will be incorporated into the schedule being used to justify the time impact.
3. Fragmentary Network (Fragnet)
 - a. Prepare a proposed fragnet for time impact analysis. The proposed fragnet shall sequence new activities into the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet shall be tied into the project schedule, including the predecessors and successors to the fragnet activities. Obtain Government approval of the proposed fragnet before incorporating it into the project schedule.
4. Time Extension
 - a. Time extensions will not be granted until after the Government has approved the Justification of Delay, including the time impact analysis. No time extension will be granted unless the delay consumes the available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.
 - b. Actual delays that the Government determines are caused by the Contractor's own actions and result in a calculated schedule delay will not be a cause for an extension to the performance period, completion date, or interim milestone date.
5. Impact to Early Completion Schedule
 - a. No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion Baseline Schedule.

3.6 SCHEDULE OF VALUES

- A. Within ten (10) days after the Contract for Construction is awarded, the Contractor shall submit to the FAA a complete itemized Schedule of Values.
- B. An approved SOV is required prior to submitting the first payment application. If there is not an approved SOV, the COR has the authority to request information to justify the costs in the payment application
- C. The schedule of values must be broken down by phase, work area, discipline, equipment, material, and labor.
- D. The Contractor's profit, fees, taxes, overhead, and General Conditions must be itemized separately and not be prorated across other categories.
- E. The Schedule of Values shall reflect accurate cost breakdowns and be supported by evidence of correctness as required by the FAA, including copies of contracts with subcontractors and material suppliers.
- F. This Schedule of Values, when approved by the FAA, shall be used to monitor the progress of the Work and as a basis for Certificates for Payment.

- G. Each item shall show its total Scheduled Value, value of previous applications, value of the application, percentage completed, value completed, and value yet to be completed. All blanks and columns must be filled in, including every percentage-complete figure.
- H. The Contractor is to add approved Change Orders to the Schedule of Values monthly.
- I. An updated schedule of values is required with each pay application

END OF SECTION 01 32 00

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

- A. Action Submittals: Written and graphic information that requires COR's responsive action.
- B. Informational Submittals: Written information that does not require COR's responsive action. Submittals may be rejected for not complying with requirements.
- C. Sustainable Design Submittals:
 - 1. Provide written and graphic information that is required to satisfy Sustainable Design documentation. Sustainable Design submittals do not require COR's responsive action. Sustainable Design submittals are those submittals indicated in individual Specification Sections as Sustainable Design submittals.
 - 2. Submittals may be rejected for not complying with requirements.
- D. Closeout Submittals:
 - 1. Provide written and graphic information that is required as a precondition to Project closeout. Closeout submittals do not require COR's responsive action. Closeout submittals are those submittals indicated in individual Specification Sections as closeout submittals.
 - 2. Submittals may be rejected for not complying with requirements.
- E. Data or Product Sheet: Supplier or manufacturer detailed information about a product, which must be included references to performance, testing, and installation requirements as required in the Specifications.
- F. Safety Data Sheets (SDS): The Contractor must submit to the COR Safety Data Sheets (SDS) for all materials and/or products utilized during the course of the project accomplishment. During the course of the project, both the COR and the Contractor must routinely check products utilized on-site to ensure only products which have had SDS submitted are utilized. Copies of all SDS must be turned over to the local FAA office for their records
- G. Shop Drawings: Fabrication drawings that show how the design will be built. Can reference both the Construction Drawings and Specifications.
- H. Samples: Representations and examples of materials to be installed, allowing for a visual verification of meeting the design intent. Electronic representations and examples are not acceptable.

- I. Certifications: Assertions of conformance with Contract requirement. (Examples include certification of adequacy of quantity of a product such as stone, certification that an item will meet code requirements, certification that an item will meet industry standards, certification of compatibility of products, etc).

1.3 SUBMITTAL PROCEDURES

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 - 1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Coordinate transmittal of different types of submittals for related parts of the Work so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. COR reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- B. Submittals Schedule: Comply with requirements in Section 01 32 00 PROJECT SCHEDULE for list of submittals and time requirements for scheduled performance of related construction activities.
- C. Processing Time: Submittals for work in the critical path shall allow enough time for submittal review, including time for resubmittals, as follows. Time for review shall commence on COR's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
 - 1. Initial Review: Allow 30 calendar days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. COR will advise Contractor when a submittal being processed must be delayed for coordination.
 - 2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
 - 3. Resubmittal Review: Allow 14 calendar days for review of each resubmittal.
- D. COR Identification: Place a permanent label or title block on each submittal for identification.
 - 1. Indicate name of firm or entity that prepared each submittal on label or title block.
 - 2. Provide a space approximately 3 by 4 inches on label or beside title block to record Contractor's review and approval markings and action taken by COR.
 - 3. Include the following information on label for processing and recording action taken:
 - a. Project name.
 - b. Date.
 - c. Name and address of Contractor.
 - d. Name and address of subcontractor.
 - e. Name and address of supplier.
 - f. Name of manufacturer.
 - g. Submittal number or other unique identifier, including revision identifier.
 - 1) Submittal number shall use Specification Section number followed by a decimal point and then a sequential number (e.g., 061000.01). Resubmittals shall include an alphabetic suffix after another decimal point (e.g., 061000.01.A).
 - h. Number and title of appropriate Specification Section.

- i. Drawing number and detail references, as appropriate.
 - j. Location(s) where product is to be installed, as appropriate.
 - k. Other necessary identification.
 - l. Where printed materials describe more than one product or model, clearly identify item to be furnished.
 - m. Show previous applicable changes made in Project by Change Orders and other instructions or remarks.
- E. Deviations: Encircle or otherwise specifically identify deviations from the Contract Documents on submittals. Indication must be reproducible when copied on black and white copier. See Section 01 25 00 SUBSTITUTION PROCEDURES.
- F. Transmittal: Package each submittal individually and appropriately for transmittal and handling. Transmit each submittal using a transmittal form. Transmittal form associated with submittals received from sources other than Contractor will be returned without review. Submittal itself may be returned at the COR discretion.
 - 1. Transmittal Form: Provide locations on form for the following information:
 - a. Project name.
 - b. Date.
 - c. Destination (To:).
 - d. Source (From:).
 - e. Names of subcontractor, manufacturer, and supplier.
 - f. Category and type of submittal.
 - g. Submittal purpose and description.
 - h. Specification Section number and title.
 - i. Drawing number and detail references, as appropriate.
 - j. Transmittal number, numbered consecutively.
 - k. Submittal and transmittal distribution record.
 - l. Remarks.
 - m. Signature of transmitter.
 - 2. On an attached separate sheet, prepared on Contractor's letterhead, record relevant information, requests for data, revisions other than those requested by COR on previous submittals, and deviations from requirements in the Contract Documents, including minor variations and limitations. Include same label information as related submittal.
- G. Resubmittals: Make resubmittals in same form as initial submittal.
 - 1. Note date and content of previous submittal.
 - 2. Note date and content of revision in label or title block and clearly indicate extent of revision with revision clouds. Failure to specifically cloud modifications shall automatically be considered cause for rejection of the modification or revision whether or not the drawing has been approved by the COR..
 - 3. Resubmit submittals until they are marked as being accepted.
 - 4. Resubmittals previously marked "no exception taken" or resubmittals not specifically marked "revise and resubmit" will not be reviewed.
- H. Request For Variations: Variations from contract requirements require Government approval pursuant to contract clause entitled "Specifications and Drawings for Construction" and will be considered where advantageous to the Government. Where variations are proposed for consideration, submit a written request, with documentation of the nature and features of the

variation and why the variation is desirable and beneficial to the Government and must include cost savings to the Government. The proposed variation shall be identified separately and included along with the required submittal for the item. When a variation is submitted for approval, the Contractor warrants the following.

1. The Contract has been reviewed to establish that the variation, when incorporated, will be compatible with other elements of the work.
 2. The Contractor shall take action and bear the additional cost, including review costs by the Government, necessary because of the proposed variation.
 3. The Contractor shall bear the cost of employing a registered professional engineer to provide stamped calculations certifying that the variation meets the project criteria.
 4. If variation involves substitution of specified product(s) or system(s) conform to requirements of Section 01 25 00 SUBSTITUTION PROCEDURES.
- I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.
- J. Use for Construction: Use only final submittals with mark indicating that no exceptions are taken by COR.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

- A. General: Prepare and submit Action Submittals required by individual Specification Sections.
1. Submit electronic (pdf) format submittals directly to the Knowledge Services Network (KSN) specifically established for Project. If files are too large for KSN, consult COR.
- B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment. Comply with Section 01 41 00 COORDINATION DRAWINGS AND BIM for incorporating approved product data as Facility Data.
1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 2. Mark each submittal to show which products and options are applicable.
 3. Include the following information, as applicable:
 - a. Manufacturer's written recommendations.
 - b. Manufacturer's product specifications.
 - c. Manufacturer's installation instructions.
 - d. Standard color charts.
 - e. Manufacturer's catalog cuts.
 - f. Wiring diagrams showing factory-installed wiring.
 - g. Printed performance curves.
 - h. Operational range diagrams.
 - i. Mill reports.
 - j. Standard product operation and maintenance manuals.
 - k. Compliance with specified referenced standards.
 - l. Testing by recognized testing agency.
 - m. Application of testing agency labels and seals.

- n. Notation of coordination requirements.
 - 4. Submit Product Data before or concurrent with Samples.
 - 5. Format: Submit Product Data in electronic (pdf) format directly to the KSN, unless otherwise indicated. COR will return marked up electronic copy and retain one copy as a Project Record Document.
- C. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data. Comply with Section 01 41 00 COORDINATION DRAWINGS AND BIM for incorporating approved shop drawing layouts into the BIM files for Coordination Drawing development.
- 1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
 - a. Dimensions.
 - b. Identification of products.
 - c. Fabrication and installation drawings.
 - d. Roughing-in and setting diagrams.
 - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring.
 - f. Shopwork manufacturing instructions.
 - g. Templates and patterns.
 - h. Schedules.
 - i. Design calculations.
 - j. Compliance with specified standards.
 - k. Notation of coordination requirements.
 - l. Notation of dimensions established by field measurement.
 - m. Relationship to adjoining construction clearly indicated.
 - n. Seal and signature of professional engineer if specified.
 - o. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring.
 - 2. Sheet Size: Except for templates, patterns, and similar full-size drawings, submit Shop Drawings in (pdf) format on sheets at least 11 by 17 inches. The resolution for full size electronic (pdf) files shall be fully legible for on screen viewing which is equivalent to a full size 22 x 34 plot.
 - 3. Format: Submit Shop drawings in electronic (pdf) format directly to the KSN, unless otherwise indicated. COR will return marked up electronic copy and retain one copy as a Project Record Document.
- D. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
- 1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 - 2. Identification: Attach label on unexposed side of Samples that includes the following:
 - a. Generic description of Sample.
 - b. Product name and name of manufacturer.
 - c. Sample source.
 - d. Number and title of appropriate Specification Section.

3. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - a. Samples that may be incorporated into the Work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
 - b. Samples not incorporated into the Work, or otherwise designated as FAA's property, are the property of Contractor.
 4. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
 - a. Number of Samples: Submit one full set of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. COR will return submittal with options selected.
 5. If sample deviates from the Specification, Contractor is responsible for detailing deviations in writing. Approval of a sample does not alleviate Contractor from Contract specifications if deviations are not noted.
- E. Product Schedule or List: As required in individual Specification Sections, prepare a written summary indicating types of products required for the Work and their intended location. Include the following information in tabular form:
1. Type of product. Include unique identifier for each product.
 2. Number and name of room or space.
 3. Location within room or space.
 4. Number of Copies: Submit two paper copies and two electronic of product schedule or list, unless otherwise indicated. COR will return one copy.
- F. Submittals required prior to Notice-to-Proceed: The following submittals must be approved by the COR prior to the contractor being allowed to proceed with the initial construction phases. No work can be done until the following submittals have been approved
1. Section 01 10 00 SUMMARY OF WORK: List of subs, certificate of insurance
 2. Section 01 32 00 PROJECT SCHEDULE
 3. Section 01 40 00 CONTRACTOR QUALITY CONTROL for CQC Plan
 4. Section 01 57 13 SOIL EROSION CONTROL
 5. Section 01 71 33 PROTECTION OF WORK AND PROPERTY
 6. Insulating Glass Unit
 7. Accident Prevention Plan.

2.2 INFORMATIONAL SUBMITTALS

- A. Prepare and submit Informational Submittals required by other Specification Sections.
1. Number of Copies: Submit electronic copies of each submittal, unless otherwise indicated.
 2. Certificates and Certifications: Provide a notarized statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.
 3. Test and Inspection Reports: Comply with requirements.

- B. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- C. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on AWS forms. Include names of firms and personnel certified.
- D. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- E. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
- F. Product Certificates: Prepare written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
- G. Material Certificates: Prepare written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.
- H. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
- I. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project. Include the following information:
 - 1. Name of evaluation organization.
 - 2. Date of evaluation.
 - 3. Time period when report is in effect.
 - 4. Product and manufacturers' names.
 - 5. Description of product.
 - 6. Test procedures and results.
 - 7. Limitations of use.
- J. Schedule of Tests and Inspections: See technical sections for require testing and inspection for construction materials and assemblies.
- K. Initial Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
- L. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before

installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.

- M. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
- N. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- O. Design Data: Prepare written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- P. Manufacturer's Instructions: Prepare written or published information that documents manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer. Include the following, as applicable:
 - 1. Preparation of substrates.
 - 2. Required substrate tolerances.
 - 3. Sequence of installation or erection.
 - 4. Required installation tolerances.
 - 5. Required adjustments.
 - 6. Recommendations for cleaning and protection.
- Q. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. Include the following, as applicable:
 - 1. Name, address, and telephone number of factory-authorized service representative making report.
 - 2. Statement on condition of substrates and their acceptability for installation of product.
 - 3. Statement that products at Project site comply with requirements.
 - 4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
 - 5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
 - 6. Statement whether conditions, products, and installation will affect warranty.
 - 7. Other required items indicated in individual Specification Sections.
- R. Safety Data Sheets (SDSs): All product submittals must be accompanied with Safety Data Sheets (SDS) from the manufacturer.

2.3 DELEGATED DESIGN

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.

1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to COR.
- B. Delegated-Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally signed PDF file paper copies of certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.
 1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these service Professional Certification:
 2. When professional certification of performance criteria of materials, systems or equipment is required by Contract Documents, COR is entitled to rely on such certifications.
 3. COR will not make independent examinations or verification of professional certifications.

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

- A. Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to COR.
- B. Submittal of shop drawings and other submittals by the Contractor shall constitute Contractor's representation that the Contractor has verified all quantities, dimensions, specified performance criteria, installation requirements, materials, catalog numbers and similar data with respect thereto and reviewed or coordinated each drawing with other Drawings and other trades.
- C. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.
- D. Ensure no work is begun until the submittals for that work have been returned with a review comment other than "Revise and Resubmit" or "Rejected".

3.2 COR'S / ACTION

- A. General: COR will not review submittals that do not bear Contractor's approval stamp and will return them without action.
- B. Action Submittals: COR will review each submittal, make marks to indicate corrections or modifications required, and return it. COR will stamp each submittal with an action stamp and will mark stamp appropriately to indicate action taken, as follows:
 1. Approved as Submitted or Approved as Noted: Proceed with fabrication, purchase, or both, of items included in submittal, subject to revisions, if any, included in COR's review comments.
 2. Not Approved: Fabrication of Work indicated cannot proceed. Correct submittal and resubmit.
 3. Revise and Resubmit:

- a. A portion of submittal is unacceptable.
 - b. Correct and resubmit the part of submittal in question.
- 4. For Information Only: No action required.
- C. The FAA may retain a construction support contractor for submittal review and other engineering support functions. The construction contractor acknowledges that no submittal, RFI, or any other information will be received directly from the construction support contractor. All information received from the construction support contractor must be validated by the FAA COR before it is considered an official response to the submittal RFI, or information inquiry.
- D. Commissioning Agent Reviews: Commissioning Agent will review submittals concurrently with Government.
- E. Informational Submittals: COR will review each submittal and will not return it, or will return it if it does not comply with requirements. COR will forward each submittal to appropriate party.
- F. Partial submittals are not acceptable, will be considered nonresponsive, and will be returned without review.
- G. Submittals not required by the Contract Documents may not be reviewed and may be discarded.
- H. The COR's review and approval of shop drawings and other submittals shall be for general conformance with the design intent of the work and with the information given in the Contract Documents only and will not in any way relieve the Contractor from:
 - 1. Conforming to the Contract Documents.
 - 2. Coordination with other trades.
 - 3. Responsibility for all required detailing and proper fitting of construction work.
 - 4. The necessity of furnishing material and workmanship required by Drawings and Specifications which may not be indicated on the shop drawings.
 - 5. Control or charge of construction means, methods, techniques, sequences or procedures, for safety precautions and programs in connection with the work.

END OF SECTION 01 33 00

SECTION 01 35 29

SAFETY AND HEALTH

PART 1 - GENERAL

1.1 SCOPE OF WORK

- A. The Contractor must control its operations and those of its Subcontractors and Suppliers to assure the least inconvenience to the traveling public. Under all circumstances, safety must be the most important consideration,
- B. Develop health, safety and emergency response procedures for construction personnel.

1.2 RELATED WORK SPECIFIED ELSEWHERE

- A. See Section 01 10 20 CONSTRUCTION FORMS for Safety and Health Checklist, FAA Life Safety System, and Pre-Construction and Maintenance Project Safety and Health Checklist (FAA 3900-18)
- B. See Section 01 33 00 SUBMITTAL PROCEDURES for submission of safety data sheets.
- C. See Section 01 77 10 FINAL CLEANING for compliance with cleaning safety procedures.

1.3 REFERENCES

- A. The publications listed below form a part of this specification to the extent referenced. The publications listed below are referenced as the latest edition published as of the date of this document. The publications are referred to within the text by the basic designation only.
 - 1. AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
 - a. ANSI/ASSP A10.32 Personal Fall Protection Systems Used in Construction and Demolition Operations
 - b. ANSI/ASSP Z359.1 Fall Protection Code
 - c. ANSI/ASSP A10.34 Protection of the Public on or Adjacent to Construction Sites
 - d. ANSI/ASSP Z359.6 Specifications and Design Requirements for Active Fall Protection Systems
 - e. ANSI/ASSP A1264.1 Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings; Stairs and Guardrail/Handrail Systems.
 - f. ANSI A14.9 American National Safety Requirements for Disappearing Attic Stairways
 - 2. ASME INTERNATIONAL (ASME)
 - a. ASME B30.3 Tower Cranes
 - b. ASME B30.22 Articulating Boom Cranes
 - c. ASME B30.5 Mobile and Locomotive Cranes
 - d. ASME B30.8 Floating Cranes and Floating Derricks
 - 3. NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
 - a. NFPA 10 Portable Fire Extinguishers

- b. NFPA 241 Safeguarding Construction, Alteration, and Demolition Operations
- c. NFPA 51B Fire Prevention During Welding, Cutting, and Other Hot Work
- d. NFPA 70 National Electrical Code
- e. NFPA 70E Electrical Safety in the Workplace
- 4. U.S. ARMY CORPS OF ENGINEERS (USACE)
 - a. COE EM 385-1-1 Safety -- Safety and Health Requirements
- 5. U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
 - a. 29 CFR 1910 Occupational Safety and Health Standards
 - b. 29 CFR 1926 Safety and Health Regulations for Construction

1.4 INFORMATION SUBMITTALS

- A. Information Submittals:
 - 1. Accident Prevention Plan (APP)
 - 2. Activity Hazard Analysis (AHA)
 - 3. Crane Critical Lift Plan
- B. Qualification data: Proof of qualification for Crane Operators
- C. Test Reports: As required by Activity Hazard Analysis.
- D. Reports:
 - 1. Accident Reports: Submit accident reports as their incidence occurs in accordance with Contractor Safety Plan.
 - 2. Scaffold inspection reports.
 - 3. Crane inspection reports.
- E. Regulatory Citations and Violations: If received from another regulatory authority, submit to COR.
- F. Certificates: Maintain on project site and make available to COR, as needed.
 - 1. Confined Space Entry Permit.
 - 2. Hot work permit
 - 3. Contractor Safety Self-Evaluation Checklist

1.5 DEFINITIONS

- A. Qualifications for Fall Protection Oversight. A competent person, as defined by OSHA, who is capable of identifying hazardous or dangerous conditions in the personal fall arrest system or any component thereof, as well as their application and use with related equipment, and has the authority to take prompt corrective measures to eliminate the hazards of falling.
- B. High Visibility Accident. Any mishap which may generate publicity and/or high visibility.
- C. Medical Treatment. Treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even though provided by a physician or registered personnel.

- D. Operating Envelope. The area surrounding any crane. Inside this "envelope" is the crane, the operator, riggers and crane walkers, rigging gear between the hook and the load, the load and the crane's supporting structure (ground, rail, etc.).
- E. Qualified Person for Fall Protection. A person with a recognized degree or professional certificate, and with extensive knowledge, training and experience in the field of fall protection; who is capable of performing design, analysis, and evaluation of fall protection systems and equipment.
- F. Recordable Injuries or Illnesses. Any work-related injury or illness that results in:
 - 1. Death, regardless of the time between the injury and death, or the length of the illness;
 - 2. Days away from work (any time lost after day of injury/illness onset);
 - 3. Restricted work;
 - 4. Transfer to another job;
 - 5. Medical treatment beyond first aid;
 - 6. Loss of consciousness; or
 - 7. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (1) through (6) above.
 - a. Weight Handling Equipment (WHE) Accident. A WHE accident occurs when any one or more of the six elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; and/or collision, including unplanned contact between the load, crane, and/or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, roll over, etc.).

1.6 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

- A. Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the pre-construction conference. The checklist will be completed monthly by the Contractor and submitted with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90, will result in a retention of up to 10 percent of the voucher.

1.7 PERSONNEL QUALIFICATIONS AND DUTIES

- A. Personnel Qualifications
 - 1. Site Safety and Health Officer (SSHO)
 - a. Site Safety and Health Officer (SSHO) shall be provided and present at the work site at all times to perform safety and occupational health management, surveillance, inspections, and safety enforcement for the Contractor. The SSHO shall be present at the project site, located so they have full mobility and reasonable access to all major work operations during the shift. An alternate SSH, shall be provided if/when the SSHO cannot be on site at a particular time. The Contractor Quality Control

(QC) person cannot be the SSHO or alternate SSHO on this project, even though the QC has safety inspection responsibilities as part of the QC duties.

- 1) The SSHO shall report to a senior project (or corporate) official.
 - 2) The SSHO, as a minimum, must produce a copy of their instructor-signed OSHA 30-hour training card (or course completion if within 90 days of having completed the training and card has not yet been issued). They will have completed:
 - a) The 30-hour OSHA General Industry safety class (may be web-based training if the student is able to directly ask questions of the instructor by chat or phone) or
 - b) The 30-hour OSHA Construction Industry safety class (may be web-based training if the student is able to directly ask questions of the instructor by chat/phone), or
 - c) As an equivalent, formal construction or industry safety and health training covering the subjects of the OSHA 30-hour course.
 - 3) SSHOs shall maintain competency through having taken 8 hours of documented formal, on-line, or self-study safety and health related coursework every year. Examples of continuing education activities that meet this requirement are writing an article, teaching a class, reading/writing professional articles, attendance/participation in professional societies/meetings, etc.
- b. Alternate SSHO:
- 1) As identified in the AHA will be assigned to insure SSHO coverage for the project at all times work activities are conducted. The Alternate SSHO must meet the same requirements and assume the responsibilities of the project SSHO. Qualifications for an Alternate SSHO shall be included on the submitted APP for approval.
 - 2) If the SSHO is off-site for a period longer than 24 hours, an Alternate SSHO shall be provided and shall fulfill the same roles and responsibilities as the primary SSHO.
2. Competent Person for Confined Space Entry:
- a. Provide a competent person for confined space meeting the definition and requirements of FAA Order 3900 19D, 29 CFR 1910.146 Permit-Required Confined Spaces and 29 CFR 1926 Subpart AA Confined Spaces in Construction.
3. Crane Operators
- a. Crane operators shall meet the requirements in USACE EM 385-1-1, Section 16 and Appendix G. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 50,000 pounds or greater, crane operators shall be designated as qualified by a source that qualifies crane operators (i.e., union, a government agency, or an organization that tests and qualifies crane operators). Proof of current qualifications shall be provided.

B. Personnel Duties:

1. Site Safety and Health Officer (SSHO)
 - a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Safety inspection logs shall be attached to the Contractors' daily quality control report.

- b. Conduct mishap investigations and complete required reports. Maintain the OSHA Form 300 and Daily Production reports for prime and sub-contractors.
 - c. Maintain applicable safety reference material on the job site.
 - d. Attend the pre-construction conference, pre-work meetings including preparatory inspection meeting, and periodic in-progress meetings.
 - e. Implement and enforce accepted APPS and AHAs.
 - f. Maintain a safety and health deficiency tracking system that monitors outstanding deficiencies until resolution. A list of unresolved safety and health deficiencies shall be posted on the safety bulletin board.
 - g. Ensure sub-contractor compliance with safety and health requirements.
2. Failure to perform the above duties will result in dismissal of the superintendent and/or SSHO, and a project work stoppage. The project work stoppage will remain in effect pending approval of a suitable replacement.

1.8 MEETINGS

- A. Pre-construction Conference:
- 1. Contractor representatives who have a responsibility or significant role in accident prevention on the project shall attend the preconstruction conference. This includes the project superintendent, site safety and health officer, quality control supervisor, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
 - 2. The Contractor shall discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the contract. This list of proposed AHAs will be reviewed at the conference and an agreement will be reached between the Contractor and the Contracting Officer's representative as to which phases will require an analysis. In addition, a schedule for the preparation, submittal, review, and acceptance of AHAs shall be established to preclude project delays.
 - 3. Deficiencies in the submitted APP will be brought to the attention of the Contractor at the preconstruction conference, and the Contractor shall revise the plan to correct deficiencies and re-submit it for acceptance. Work shall not begin until there is an accepted APP.
 - 4. The functions of the Preconstruction conference may take place at the Post-Award Kickoff meeting for Design Build Contracts.
- B. Safety Meetings: Minutes showing contract title, signatures of attendees and a list of topics discussed shall be attached to the Contractors' daily report.

1.9 ACCIDENT PREVENTION PLAN (APP)

- A. The Contractor shall use a qualified person to prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of USACE EM 385-1-1 and as supplemented herein.
- B. Cover all paragraph and subparagraph elements in USACE EM 385-1-1, Appendix A, "Minimum Basic Outline for Accident Prevention Plan". Specific requirements for some of the APP elements are described below.

1. The APP shall be job-specific and shall address any unusual or unique aspects of the project or activity for which it is written.
 2. The APP shall interface with the Contractor's overall safety and health program.
 3. Any portions of the Contractor's overall safety and health program referenced in the APP shall be included in the applicable APP element and made site-specific.
 4. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out.
 5. The APP shall be signed by the person and firm (senior person) preparing the APP, the Contractor, the on-site superintendent, the designated site safety and health officer and any designated CSP and/or CIH.
- C. Submit the APP to the Contracting Officer 30 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP.
- D. Once accepted by the Contracting Officer, the APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted APP will be cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified.
- E. Once work begins, changes to the accepted APP shall be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and quality control manager. Should any hazard become evident, stop work in the area, secure the area, and develop a plan to remove the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate/remove the hazard. In the interim, all necessary action shall be taken to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSE A10.34,) and the environment.
- F. Copies of the accepted plan will be maintained at the resident engineer's office and at the job site.
- G. The APP shall be continuously reviewed and amended, as necessary, throughout the life of the contract. Unusual or high-hazard activities not identified in the original APP shall be incorporated in the plan as they are discovered.
- 1.10 ACTIVITY HAZARD ANALYSIS (AHA)
- A. The Activity Hazard Analysis (AHA) format shall be in accordance with USACE EM 385-1-1. Submit the AHA for review at least 15 calendar days prior to the start of each phase. Format subsequent AHAs as amendments to the APP. The analysis should be used during daily inspections to ensure the implementation and effectiveness of the activity's safety and health controls.
- B. The AHA list will be reviewed periodically (at least monthly) at the Contractor supervisory safety meeting and updated as necessary when procedures, scheduling, or hazards change.
- C. The activity hazard analyses shall be developed using the project schedule as the basis for the activities performed. Any activities listed on the project schedule will require an AHA. The AHAs

will be developed by the contractor, supplier or subcontractor and provided to the prime contractor for submittal to the Contracting Officer.

1.11 DISPLAY OF SAFETY INFORMATION

- A. Within one calendar days after commencement of work, erect a safety bulletin board at the job site. The safety bulletin board shall include information and be maintained as required by USACE EM 385-1-1, section 01. A07. Additional items required to be posted include:
 - 1. Confined space entry permit.
 - 2. Hot work permit.

1.12 SITE SAFETY REFERENCE MATERIALS

- A. Maintain safety-related references applicable to the project, including those listed in the article "References". Maintain applicable equipment manufacturer's manuals.

1.13 EMERGENCY MEDICAL TREATMENT

- A. Contractors will arrange for their own emergency medical treatment. Government has no responsibility to provide emergency medical treatment.

1.14 SAFETY AND HEALTH PROGRAM

- A. Contractor Responsibility: The Contractor shall assume full responsibility and liability for compliance with applicable codes, standards and regulations pertaining to the health and safety of personnel during execution of the Work.
- B. First Aid and Emergency Response Requirements: The Contractor shall provide for emergency first aid equipment. Additionally, a 20-pound ABC-rated fire extinguisher shall be maintained on-site as well as absorbent material of sufficient quantity to collect any spill which might occur during this project. A listing of emergency phone numbers and points of contact for fire, hospital, police, ambulance, and other necessary contacts shall be posted at the Contractor's site.
- C. Contractor shall provide for construction site visitors Personal Protective Equipment (PPE) per OSHA for use during their visits. Provide a minimum of 10 sets with replacements for items not suitable for re-use.

1.15 REPORTS

- A. Accident Reports:
 - 1. For recordable injuries and illnesses, and property damage accidents resulting in at least \$2,000 in damages, the Prime Contractor shall conduct an accident investigation to establish the root cause(s) of the accident, complete the Navy Contractor Significant Incident Report (CSIR) USACE Accident Report Form 3394 and provide the report to the Contracting Officer within 5 calendar day(s) of the accident. The Contracting Officer will provide copies of any required or special forms.
 - 2. For any weight handling equipment accident (including rigging gear accidents) the Prime Contractor shall conduct an accident investigation to establish the root cause(s) of the accident, complete the WHE Accident Report (Crane and Rigging Gear) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Crane

operations shall not proceed until cause is determined and corrective actions have been implemented to the satisfaction of the contracting officer. The Contracting Officer will provide a blank copy of the accident report form.

B. Accident Notification:

1. Notify the Contracting Officer as soon as practical, but not later than four hours, after any accident meeting the definition of Recordable Injuries or Illnesses or High Visibility Accidents, property damage equal to or greater than \$2,000, or any weight handling equipment accident. Information shall include contractor name; contract title; type of contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (to include type of construction equipment used, PPE used, etc.). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted.

C. Crane Reports

1. Submit crane inspection reports required in accordance with USACE EM 385-1-1, Appendix H and as specified herein with Daily Reports of Inspections.

D. Certificate of Compliance:

1. The Contractor shall provide a Certificate of Compliance for each crane entering an activity under this contract (see Contracting Officer for a blank certificate). Certificate shall state that the crane and rigging gear meet applicable OSHA regulations (with the Contractor citing which OSHA regulations are applicable, e.g., cranes used in construction, demolition, or maintenance shall comply with 29 CFR 1926 and USACE EM 385-1-1 section 16 and Appendix H. Certify on the Certificate of Compliance that the crane operator(s) is qualified and trained in the operation of the crane to be used. The Contractor shall also certify that all of its crane operators working on site have been trained in the proper use of all safety devices (e.g., anti-two block devices). These certifications shall be posted on the crane.

1.16 LOCKOUT/TAG OUT PROCEDURES

A. Prepare written Lockout/Tagout (LO/TO) procedures, as needed by OSHA, for construction, with detailed steps (including photos) for shutting down and isolating energy sources on machines or equipment to control hazardous energy.

1. Include procedural steps for placement, removal, and transfer of lockout/tagout devices.
2. Verify that isolation and de-energization of the equipment is accomplished before starting work on locked-out/tagged-out machines or equipment (when verifying de-energization, ensure the multimeter is operating properly).

B. Submit a copy of the Hazardous Energy Control Program for review to the FAA site representative (e.g., Contracting Officer Representative, COR), prior to starting work. As part of the Hazardous Energy Control Program, Contractor must provide their own LO/TO equipment and ensure all Contractor personnel have been trained.

1.17 FALL HAZARD PREVENTION PROGRAM

- A. Scaffolds: A competent person shall delineate the fall protection requirements necessary during the erection and dismantling operation of scaffolds used on the project in the fall protection plan and activity hazard analysis for the phase of work.
- B. Training: A competent person shall institute a fall protection program. As part of the Fall Protection Program, Contractor shall provide training for each employee who might be exposed to fall hazards

1.18 DRUG PREVENTION PROGRAM

- A. Conduct a proactive drug and alcohol use prevention program for all workers, prime and subcontractor, on the site. Ensure that no employees either use illegal drugs or consume alcohol during work hours. Ensure that no employees are under the influence of drugs or alcohol during work hours. After accidents, collect blood, urine or saliva specimens and test injured employee for influence or drugs or alcohol. A copy of the test shall be made available to the Resident Engineer upon request.

1.19 HIGH HAZARD WORK AND LONG DURATION

- A. Work under this contract is potentially hazardous. Pursuant to contract clause “AMS 52.236-13, Accident Prevention, Alternate I,” submit in writing additional proposals for effecting accident prevention under hazardous conditions. Meet in conference with COR to discuss and develop mutual understanding relative to the administration of the overall safety program.

1.20 FIRE-SAFETY PROGRAM:

- A. Show compliance with requirements of NFPA 241 and authorities having jurisdiction. Indicate Contractor personnel responsible for management of fire-prevention program. Deviations from temporary egress plans must be approved by COR.

PART 2 - PRODUCTS

2.1 FALL PROTECTION ANCHORAGE

- A. Fall protection anchorages, used by contractors to protect their people, will be left in place and so identified for continued customer use. COR will make a determination as to the left in place of anchorages.

2.2 CONFINED SPACE SIGNAGE

- A. Provide permanent signs integral to or securely attached to access covers for new confined spaces. Signs wording: “DANGER—PERMIT REQUIRED CONFINED SPACE – DO NOT ENTER”

2.3 JOBSITE SAFETY

- A. Comply with SSHO and Jobsite Safety Plan requirements specified in this Section.

- B. Emergency Assistance: The Contractor shall post, at the site, telephone numbers for reporting emergencies, including ambulance, police, fire department, gas utility, electric utility, water/sewer utility, poison prevention aid and hazardous-waste handling. This information shall be posted in a conspicuous location within the project area prior to the start of any work at the site.
- C. Safety Signs: The Contractor shall post legible accident prevention signs in construction areas in accordance with OSHA standards. Safety signs shall conform to ANSI 235.1 and 235.2 Vehicular traffic control devices, barricades and signals shall conform to ANSI D6.1.
- D. Emergency Evacuation: The Contractor shall post evacuation routes and facility emergency/self-protection plans at the site, train all employees in emergency procedures and document such training. In the event of a fire, the Contractor shall immediately activate the alarm at the nearest fire alarm pull station and notify building security. Upon the activation of the audible alarm, the building will be evacuated. No personnel shall reenter the facility until security personnel signal that the building is safe.
- E. Contractor Personnel to be Contacted: The Contractor shall submit a written list of emergency telephone numbers and names of persons to contact for the General Contractor superintendent and for each major sub-contractor working on the project site. The initial list shall be submitted to the COR at the Preconstruction Meeting. The list shall be updated and resubmitted to the COR as needed.

PART 3 - EXECUTION

3.1 CONSTRUCTION

- A. Comply with NFPA 241, the accident prevention plan, the activity hazard analysis and other related submittals and activity fire and safety regulations.
- B. Hazardous Material Exclusions: Notwithstanding any other hazardous material used in this contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation as well as materials which contain asbestos, mercury or polychlorinated biphenyls, diisocyanates, lead-based paint are prohibited. Exceptions to the use of any of the above excluded materials may be considered by COR upon written request by Contractor.
- C. Unforeseen Hazardous Material: If material that may be hazardous to human health upon disturbance during construction operations is encountered, stop that portion of work and notify the COR immediately. Within 14 calendar days the COR will determine if the material is hazardous. If material is not hazardous or poses no danger, the COR will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the COR will issue a modification pursuant to "AMS 52.243-4, Changes" and "AMS 52.236-2, Differing Site Conditions".

3.2 PRE-OUTAGE COORDINATION MEETING

- A. Contractors are required to apply for utility outages a minimum of 15 days in advance. As a minimum, the request should include the location of the outage, utilities being affected, duration of outage and any necessary sketches. Once approved and prior to beginning work on the utility system

requiring shut down, the Contractor shall attend a pre-outage coordination meeting with the COR to review the scope of work and the lock out/tag out procedures for work protection.

3.3 PERSONNEL PROTECTION

A. Personnel Protective Equipment:

1. Special facilities, devices, equipment and similar items used by the Contractor in execution of the Work shall comply with the applicable regulations.
2. Personal protective equipment for eyes, face, ears, nose, head, extremities and/or full body shall be provided, used and properly maintained by the Contractor whenever necessitated by reasons of hazards encountered in a manner capable of causing illness, injury or impairment in the function of any part of the body.
3. Persons required to use personal protective equipment shall be thoroughly trained. Training programs shall, as a minimum, meet OSHA and EPA requirements where applicable. The Contractor shall submit proof and criteria for employee training as requested.

B. Hazardous Noise: Provide hazardous noise signs, and hearing protection, wherever equipment and work procedures produce sound-pressure levels greater than 85 dBA steady state or 140 dBA impulse, regardless of the duration of the exposure.

C. Fall Protection: Enforce use of the fall protection device named for each activity in the AHA all times when an employee is on a surface 4 feet or more above lower levels. Personal fall arrest systems are required when working from an articulating or extendible boom, scissor lifts, swing stages, or suspended platform. Fall protection must comply with ANSI A10.32.

1. Personal Fall Arrest Device: Equipment, subsystems and components shall meet ANSI Z359.1 Fall Protection Code.. Only with a shock absorbing lanyard is an acceptable personal fall arrest device. Full Body Harness may only be used as positioning devices only such as for steel reinforcing assembly. Body belts are not authorized as a personal fall arrest device. Harnesses must have upper middle back "D" rings for proper body suspension during a fall. Lanyard must be fitted with a double locking snap hook attachment. Webbing, straps, and ropes must be of synthetic fiber or wire rope.
2. Fall Protection for Roofs:
 - a. For work within 6 feet of an edge, on low pitched roofs, personnel shall be protected by use of personal fall arrest systems, guardrails, safety nets. Safety monitoring system is not adequate fall protection and is not authorized.
 - b. For work greater than 6 feet from an edge, warning lines shall be erected and installed in accordance with 29 CFR 1926.502(f).
3. Safety Nets: Safety nets shall be provided in unguarded workplaces more than 25 feet above surface.

D. Scaffolding: Employees shall be provided with safe means of access to the work area on the scaffold. Climbing on any scaffold braces or supports not specifically designed for access is prohibited. Contractor shall ensure that scaffold erection is performed by employees that are qualified. Do not use scaffold without the capability of supporting at least four times the maximum intended load or without appropriate fall protection as delineated in the accepted fall protection plan. Minimum platform size shall be based on the platform not being greater in height than four times the dimension of the smallest width dimension for rolling scaffold. Some Baker type scaffolding has been found not to meet these requirements. Stationary scaffolds must be

attached to structural building components to safeguard against tipping forward or backward. The first tie-in shall be at the height equal to 4 times the width of the scaffold base.

- E. Excavations: A competent person for excavation shall be on site when work is being performed in excavation and shall inspect excavations prior to entry by workers. Individuals must evaluate for all hazards, including atmospheric, necessary to correct hazards promptly.
- F. Conduct of Electrical Work: Underground electrical spaces must be certified safe for entry before entering to conduct work. Cable intended to be cut must be positively identified and de-energized prior to performing each cut. Perform all high voltage cutting remotely. When racking in or live switching of circuit breakers, no additional person other than the switch operator will be allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method. When working in energized substations, only qualified electrical workers shall be permitted to enter. When work requires Contractor to work near energized circuits as defined by the NFPA 70, high voltage personnel must use personnel protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves with leather protective sleeves, fire retarding shirts, coveralls, face shields, and safety glasses. Insulating blankets, hearing protection, and switching suits may be required, depending on the specific job and as delineated in the Contractor AHA.
- G. Work in Manholes: Contractor shall provide mechanical ventilation for all work accomplished in manholes, unless other hazards are present like friable asbestos.
- H. Work in Confined Spaces: Comply with the requirements of FAA Order 3900.19D, 29 CFR 1910.146 Permit-required confined spaces, and 29 CFR 1926 Subpart AA Confined Spaces in Construction. Any potential for a hazard in the confined space requires a permit system to be used.
 - 1. Entry Procedures. Prohibit entry into a confined space by personnel for any purpose, including hot work, until the competent person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. All space hazards shall be reviewed with each employee during review of the AHA.
 - 2. Forced air ventilation is required for all permit-required confined space entry operations and the minimum air exchange requirements must be maintained.
 - 3. Train all employees involved in permit-required confined space entry.
 - 4. Entry Permit. Completed prior to any permit-required confined space entry.
- I. Crystalline Silica: Grinding, abrasive blasting, and foundry operations of construction materials containing crystalline silica, shall comply with OSHA regulations, 1926.1153 - Respirable crystalline silica and COE EM-385-1-1, (Appendix C). The Contractor shall develop and implement effective exposure control and elimination procedures to include dust control systems, engineering controls, and establishment of work area boundaries, as well as medical surveillance, training, air monitoring, and personal protective equipment.

3.4 ACCIDENT SCENE PRESERVATION

- A. For serious accidents, ensure the accident site is secured and evidence is protected remaining undisturbed until released by the COR. After release is issued, promptly replace used, damaged, or worn equipment.

3.5 EMERGENCY SUSPENSION OF WORK

- A. When the Contractor is notified by COR of non-compliance with the safety or health provisions of the Contract, the Contractor shall immediately cease work in the subject area unless otherwise instructed and correct the unsafe or unhealthy condition.
 - 1. If the Contractor fails to comply promptly, all or part of the Work will be stopped by notice from COR.
 - 2. When COR determines that satisfactory corrective action has been taken by the Contractor, work shall resume.
 - 3. The Contractor shall not be allowed any extension of time or compensation for damages in connection with a work stoppage for an unsafe or unhealthy condition.

3.6 EQUIPMENT

- A. Material Handling Equipment
 - 1. Material handling equipment such as forklifts shall not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions.
 - 2. The use of hooks on equipment for lifting material must be in accordance with manufacturer's printed instructions.
 - 3. Operators of forklifts or power industrial trucks must ensure that each powered industrial truck operator is competent to operate a powered industrial truck safely, as demonstrated by the successful completion of the training and evaluation in accordance with OSHA 29 CFR 1910.178(L).

3.7 EXCAVATIONS

- A. The competent person obtain soil report to determine soil classification..
- B. Utility Locations
- C. Excavations: The competent person for excavation shall be on site when work is being performed excavation, and shall inspect excavations prior to entry by workers. Individual must evaluate for all hazards, including atmospheric, necessary to correct hazards promptly. Prior to digging, the appropriate digging permit must be obtained. All underground utilities in the work area must be positively identified by a private utility locating service in addition to any station locating service and coordinated with the station utility department. Any markings made during the utility investigation must be maintained throughout the contract.

END OF SECTION 01 35 29

SECTION 01 40 00

CONTRACTOR QUALITY CONTROL

PART 1 - GENERAL

1.1 SUPERVISION AND CONSTRUCTION PROCEDURES

- A. At all times during performance of this contract, and until the work is completed and accepted, the Contractor must directly supervise the work or assign and have on the worksite a competent superintendent who is an employee of the Contractor and is satisfactory to the CO and has the authority to act for the Contractor.
- B. The Contractor must supervise and direct the Work, using the Contractor's best skill and attention. The Contractor is solely responsible for and has control over construction means, methods, techniques, sequences, and procedures and for coordinating all portions of the Work under the Contract including coordination of the duties of all trades, unless the Contract Documents give other specific instructions concerning these matters.
- C. The Contractor must control its operations and those of its Subcontractors and Suppliers to assure the least inconvenience to the traveling public. Under all circumstances, safety must be the most important consideration.
- D. Contractor must lay out all work well enough in advance to avoid conflicts or interferences with other work in progress so that in case of interference the layout may be altered to suit the conditions, prior to the installation of any work and without additional cost to the FAA. The Contractor must be responsible to coordinate all work and take all action as required to avoid conflicts between trades.

1.2 REFERENCES

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.
 - 1. AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
 - a. ASTM D 3740 Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
 - b. ASTM E 329 Agencies Engaged in Construction Inspection, Testing, or Special Inspection.

1.3 ACTION SUBMITTALS

- A. Contractor Quality Control Plan: As specified herein.

1.4 INFORMATION SUBMITTALS

- A. Contractor Quality Control Personnel: Submit list of personnel and their qualifications.
- B. Daily Logs: Submit daily CQC reports.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

3.1 QUALITY CONTROL PROCESS AND PERSONNEL

- A. The Contractor is responsible for quality control and must establish and maintain an effective quality control system. The quality control system must consist of plans, procedures, and organization necessary to produce an end product that complies with the contract requirements.
 - 1. The system must cover all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence.
 - 2. The site project superintendent will be held responsible for the quality of work on the job and is subject to removal by the COR for non-compliance with the quality requirements specified in the contract. The site project superintendent in this context must be the highest level manager responsible for the overall construction activities at the site, including quality and production. The site project superintendent must maintain a physical presence at the site at all times, except as otherwise acceptable to the COR, and must be responsible for all construction and construction related activities at the site.
 - 3. Similar requirements apply to the quality control manager.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

- A. Content of the CQC Plan:
 - 1. Plan shall be submitted no later than 15 days after Notice to Proceed (NTP). Reviewing procedures will be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES
 - 2. The CQC Plan must include, as a minimum, the following to cover all construction operations, both onsite and offsite, including work by subcontractors, fabricators, suppliers, and purchasing agents:
 - a. A description of the quality control organization, including a chart showing lines of authority.
 - b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.
 - c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authority to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the contract. The CQC System Manager must issue letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities. Copies of these letters must also be furnished to the Government.
 - d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. These procedures must be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.
 - e. Control, verification, and acceptance testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test

- frequency, and person responsible for each test. (Laboratory facilities must be approved by the COR.)
- f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.
 - g. Procedures for tracking construction deficiencies from identification through acceptable corrective action. These procedures must establish verification that identified deficiencies have been corrected.
 - h. Reporting procedures, including proposed reporting formats.
 - i. A list of the definable features of work. A definable feature of work is a task that is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or it may be work by the same trade in a different environment. Although each section of the specifications may generally be considered as a definable feature of work, there is frequently more than one definable feature under a particular section. This list will be agreed upon during the coordination meeting.
- 3. Acceptance of Plan: Acceptance of the Contractor's plan is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes in his CQC Plan and operations including removal of personnel, as necessary, to obtain the quality specified.
 - 4. Notification of Changes: After acceptance of the CQC Plan, the Contractor must notify the COR in writing of any proposed change. Proposed changes are subject to acceptance by the COR.

3.3 COORDINATION MEETING

- A. After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, the Contractor must meet with the COR and discuss the Contractor's quality control system. The CQC Plan must be submitted for review a minimum of 7 calendar days prior to the Coordination Meeting. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the FAA's Quality Assurance. Minutes of the meeting will be prepared by the Contractor and signed by both the Contractor and the COR.. There may be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings and/or address deficiencies in the CQC system or procedures that may require corrective action by the Contractor.

3.4 INDEPENDENT QUALITY CONTROL ORGANIZATION (QC)

- A. Personnel Requirements: See Section 01 31 00 CONTRACTOR QUALITY CONTROL for requirements for the Independent QC (IQC) Manager.
 - 1. Include additional qualified personnel as determined by the IQC Manager. contract compliance.
 - 2. The IQC staff must maintain a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure contract compliance. The IQC staff must be subject to acceptance by the COR.
 - 3. The Contractor must provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC and IQC organization.

4. Complete records of all letters, material submittals, shop drawing submittals, schedules and all other project documentation must be promptly furnished to the CQC organization by the Contractor. The CQC organization must be responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the COR.

B. IQC Manager Requirements:

1. Attend all weekly and coordination meetings as required by the project.
2. Coordinate quality control for all disciplines as required per the contract documents.
3. Interact directly with the Contractor's QC manager and with the Government to implement the QC plan.
4. An alternate for the IQC Manager must be identified in the plan to serve in the event of the IQC Manager's absence. The requirements for the alternate must be the same as for the designated IQC Manager.

3.5 CONTROL

- A. Contractor Quality Control is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control must be conducted by the IQC Manager for each definable feature of work as follows:

- B. Preparatory Phase: This phase must be performed prior to beginning work on each definable feature of work; after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase must include:

1. A review of each paragraph of applicable specifications, reference codes, and standards. A copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field must be made available by the Contractor at the preparatory inspection. These copies must be maintained in the field and available for use by Government personnel until final acceptance of the work.
2. A review of the contract drawings.
3. A check to assure that all materials and/or equipment have been tested, submitted, and approved.
4. Review of provisions that have been made to provide required control inspection and testing.
5. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the contract.
6. A physical examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.
7. A review of the appropriate activity hazard analysis to assure safety requirements are met.
8. Discussion of procedures for controlling quality of the work including repetitive deficiencies.
9. Document construction tolerances and workmanship standards for that feature of work.
10. A check to ensure that the portion of the plan for the work to be performed has been accepted by the COR.
11. Discussion of the initial control phase.
12. The Government must be notified at least 48 hours in advance of beginning the preparatory control phase. This phase must include a meeting conducted by the IQC Manager and attended by the superintendent, other IQC personnel (as applicable), and the foreman responsible for the definable feature. The results of the preparatory phase actions must be documented by separate minutes prepared by the IQC Manager and attached to the daily IQC

report. The Contractor must instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.

- C. Initial Phase: This phase must be accomplished at the beginning of a definable feature of work. The following must be accomplished:
 - 1. A check of work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.
 - 2. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing.
 - 3. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
 - 4. Resolve all differences.
 - 5. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
 - 6. The Government must be notified at least 24 hours in advance of beginning the initial phase. Separate minutes of this phase must be prepared by the IQC Manager and attached to the daily CQC report. Exact location of initial phase must be indicated for future reference and comparison with follow-up phases.
 - 7. The initial phase should be repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.
- D. Follow-up Phase: Daily checks must be performed to assure control activities, including control testing, are providing continued compliance with contract requirements, until completion of the particular feature of work. The checks must be made a matter of record in the CQC documentation. Final follow-up checks must be conducted and all deficiencies corrected prior to the start of additional features of work that may be affected by the deficient work. The Contractor must not build upon nor conceal non-conforming work.
- E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases must be conducted on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.6 TESTS

- A. Testing Procedures: The Contractor must perform specified or required tests to verify that control measures are adequate to provide a product that conforms to contract requirements. Upon request, the Contractor must furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and/or acceptance tests when specified. The Contractor must procure the services of an approved testing laboratory or establish an approved testing laboratory at the project site. The Contractor must perform the following activities and record and provide the following data:
 - 1. Verify that testing procedures comply with contract requirements.
 - 2. Verify that facilities and testing equipment are available and comply with testing standards.
 - 3. Check test instrument calibration data against certified standards.
 - 4. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.

5. Results of all tests taken, both passing and failing tests, must be recorded on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test must be given. If approved by the COR, actual test reports may be submitted later with a reference to the test number and date taken. An information copy of tests performed by an offsite or commercial test facility must be provided directly to the COR. Failure to submit timely test reports as stated may result in nonpayment for related work performed and disapproval of the test facility for this contract.
- B. See Section 01 45 23 TESTING AND INSPECTING SERVICES for third party testing agency requirements and procedures.
- C. Coordination for each specific test, exact delivery location, and dates will be made through the Contracting Officer's Representative.

3.7 DOCUMENTATION

- A. The Contractor must maintain current records providing factual evidence that required quality control activities and/or tests have been performed. These records must include the work of subcontractors and suppliers and must be on an acceptable form that includes, as a minimum, the following information:
 1. Contractor/subcontractor and their area of responsibility.
 2. Operating plant/equipment with hours worked, idle, or down for repair.
 3. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.
 4. Test and/or control activities performed with results and references to specifications/drawings requirements. The control phase must be identified (Preparatory, Initial, and Follow-up). List of deficiencies noted, along with corrective action.
 5. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.
 6. Submittals and deliverables reviewed, with contract reference, by whom, and action taken.
 7. Offsite surveillance activities, including actions taken.
 8. Instructions given/received and conflicts in plans and/or specifications.
 9. Contractor's verification statement.
- B. These records must indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered.
 1. These records must cover both conforming and deficient features and must include a statement that equipment and materials incorporated in the work and workmanship comply with the contract.
 2. The records in report form must be furnished to the Government daily within 24 hours after the date covered by the report, except that reports need not be submitted for days on which no work is performed.
 3. As a minimum, one report must be prepared and submitted for every 7 days of no work and on the last day of a no work period.
 4. All calendar days must be accounted for throughout the life of the contract. The first report following a day of no work must be for that day only.
 5. Reports must be signed and dated by the CQC System Manager.

6. The report from the CQC System Manager must include copies of test reports and copies of reports prepared by all subordinate quality control personnel.

3.8 NOTIFICATION OF NONCOMPLIANCE

- A. The COR will notify the Contractor of any detected noncompliance with the foregoing requirements. The Contractor must take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, must be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the COR may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders must be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

END OF SECTION 01 40 00

SECTION 01 45 23

TESTING AND INSPECTING SERVICES

PART 1 - GENERAL

1.1 SUMMARY

- A. Description of Work: Work of this Section includes, but is not limited to, the following:
 - 1. Independent testing and inspection agency responsibilities and requirements for quality control testing and inspection.
 - 2. Contractor responsibilities for quality control testing and inspection.
 - 3. Requirements for reports by independent agency.
- B. General:
 - 1. Services of independent testing and inspection agencies will be used by Owner to determine whether the Work is in general compliance with requirements. Tests and inspections that demonstrate compliance of a portion of the Work with the Contract Documents, shall not be construed to indicate that untested or uninspected portions of the Work are free from defects.
 - 2. Contract Work shall conform to indicated requirements. Nothing shall be construed to modify this responsibility, except for explicit acceptance by Architect and Owner of specific items that do not meet requirements.
- C. Work Included: Scope and extent of quality control testing and inspection by independent agency is specified in technical Specification Sections.
 - 1. In addition, provide special inspections as required by Building Code and authorities having jurisdiction (AHJ). Fill out applicable forms required by AHJ.
- D. Payment:
 - 1. Contractor is responsible for expense of:
 - a. Testing and inspection of Work not evidencing compliance with Contract Documents.
 - b. Testing and inspection to verify adequacy of Work done without prior notice, without proper supervision, or contrary to standard construction practices.
 - c. Correction of Work not evidencing compliance with Contract Documents, including re-inspection or re-testing required to show compliance.
 - d. Additional testing and inspection required due to changes made or proposed by the Contractor.
 - e. Overtime work.
 - f. Punch list work.
 - g. Testing/inspection trips canceled without minimum of 12 hours' notice.
 - h. Schedule delay, including waiting time at Project site in excess of 1 hour when Work is not ready for inspection.
 - 2. Contractor shall engage and pay for independent agency services required by Contract Documents.

1.2 INFORMATIONAL SUBMITTALS

- A. Tests and Inspections Summary:
1. Contractor shall submit summary and schedule of laboratory and field quality control tests and inspections required by the Contract Documents.
 2. Submit at least 7 days prior to first scheduled test or inspection.
 3. Schedule of Tests and Inspections: Prepare in tabular form and include the following:
 - a. Specification Section number and title.
 - b. Entity responsible for performing tests and inspections.
 - c. Description of test and inspection.
 - d. Identification of applicable standards.
 - e. Identification of test and inspection methods.
 - f. Number of tests and inspections required.
 - g. Time schedule or time span for tests and inspections.
 - h. Requirements for obtaining samples.
 - i. Unique characteristics of each quality-control service.
- B. Testing and Inspection Reports - Independent Agency:
1. Submit reports immediately following completion of test or inspection.
 2. Submit weekly reports of on-site observations.
 3. Distribute to Architect, Owner, Contractor, and AHJ (as required or requested by AHJ).
- C. Qualification Data Required from Independent Agency:
1. Submit copy of report of inspection of independent agencies' testing and laboratory facilities made by the Materials Reference Laboratory of National Institute of Standards and Technology during most recent tour of inspection, including memorandum of remedies of deficiencies reported by inspection, if any.
 2. Submit Agency's permits, licenses, and certificates.
 3. Submit verification of required insurance coverage.
 4. Submit documentation of personnel qualifications.

1.3 QUALITY ASSURANCE

- A. Testing and Inspecting Agency Qualifications: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspection indicated, as documented in accordance with ASTM E329, and with additional qualifications specified in individual Sections; and, where required by authorities having jurisdiction, that is acceptable to authorities.
1. Independent Agency Qualifications:
 - a. Licensed to operate in Project jurisdiction.
 - b. Perform inspection and testing services under supervision of a Registered Professional Engineer, licensed in Project jurisdiction.
- B. For personnel responsible for inspection, sampling or testing concrete: Certified by American Concrete Institute as a Concrete Field-Testing Technician, Grade I.
- C. Calibration of Equipment: Calibrate at maximum 12-month intervals by devices of accuracy traceable to either National Institute of Standards or accepted values of natural physical constants.

D. Test Results:

1. If test samples fail to meet requirements of Contract Documents, materials or equipment shall be subject to removal and replacement by Contractor with materials or equipment meeting requirements of Contract Document, or at discretion of Owner, installed defective material and equipment may be permitted to remain in place subject to adjustment of Contract Sum.
2. If initial tests are defective, costs of retesting shall be borne by Independent Agency.

PART 2 - PRODUCTS

2.1 TEST SAMPLES

- A. Samples of actual materials or equipment to be used in completed Work as provided by Contractor and delivered by Contractor to agency.

PART 3 - EXECUTION

3.1 TESTING AND INSPECTION

A. Independent Agency Responsibilities:

1. Conduct and interpret specified inspections and tests and report whether material inspected and specimens tested comply with requirements of Contract Documents.
2. Evaluate adequacy of Contractor's quality control procedures.
3. Cooperate with Architect, Owner, and Contractor.
4. Provide qualified personnel.
5. Promptly notify Architect, Owner and Contractor of observed irregularities or deficiencies of Work or products used in Work.
6. Submit written reports of each test and inspection within 24 hours of each test and inspection.
7. Submit daily reports of on-site observations, unless otherwise specified.
8. Perform additional tests and inspections at request of Architect or Owner.
9. Attend Project progress meetings as requested by Architect or Owner.
10. Submit monthly summary.
11. Submit final full project report by permit number.

B. Limitations of Authority of Independent Agency:

1. Agency is not authorized to:
 - a. Release, revoke, alter or enlarge requirements of Contract Documents.
 - b. Approve, accept, or reject Work.
 - c. Perform duties of Contractor.

C. Test and Inspection Reports: Prepare and submit certified written reports specified in other Sections. Include the following:

1. Date of issue.
2. Project title and number.
3. Name, address, telephone number, and email address of testing agency.
4. Dates and locations of samples and tests or inspections.
5. Names of individuals making tests and inspections.

6. Description of the Work and test and inspection method.
7. Identification of product and Specification Section.
8. Complete test or inspection data.
9. Test and inspection results and an interpretation of test results.
10. Record of temperature and weather conditions at time of sample-taking and testing and inspection.
11. Comments or professional opinion on whether tested or inspected Work complies with the Contract Document requirements.
12. Name and signature of laboratory inspector.
13. Recommendations on retesting and reinspecting.

D. Contractor's Responsibilities:

1. Provide independent agencies with at least 24 hours advance notice of time when Work which requires testing or inspection will be performed.
2. Cooperate with agency personnel, provide access to Work and to manufacturer's operations where applicable.
3. Secure and deliver to agency adequate quantities of representative samples of materials proposed for use which require testing.
4. Furnish copies of manufacturer's test reports of products as required.
5. Ensure Work to be inspected has been completed and inspected by Contractor prior to independent agency inspection.
6. Furnish incidental labor as required to facilitate work of independent agency, including the following:
 - a. Provide access to Work.
 - b. Obtain and handle samples.
 - c. Facilitate inspections and tests.
7. Make arrangements with agency and pay for additional samples, tests and inspections required when initial tests indicate Work does not conform to Contract Documents.
8. Testing and inspection does not relieve Contractor from fulfilling requirements of Contract Documents.

3.2 TESTING REQUIREMENTS

- A. See technical sections for testing requirements of products and assemblies.

END OF SECTION 01 45 23

SECTION 01 47 00

SECURITY REQUIREMENTS

PART 1 - GENERAL

1.1 GENERAL SECURITY REQUIREMENTS

- A. The Contractor and his employees must comply with security requirements imposed by the FAA, including any necessary security clearances. Failure to inspect the site or obtain knowledge of security regulations shall not relieve the Contractor from security requirements or from performance of any part of the work. In addition to adhering to FAA requirements, the Contractor shall submit a comprehensive Security Plan to the COR for approval prior to the start of Work. Plans shall provide site and building security plans as well as security procedures related to the sequence of Work.
- B. Prior to the start of work on the site, the Contractor shall submit, to the COR for approval, a list of the names, last four digits of social security numbers and addresses of all employees and subcontractor employees. The list shall identify the Prime Contractor and each subcontractor and trade. It shall be updated, as necessary, to accurately identify all workers who will be working on the site during the project.
- C. The name and telephone number of the Contractor's Superintendent and authorized alternate individual who can be reached on a 24-hour basis shall be provided to the COR at the Preconstruction Meeting.
- D. Emergency and Other Contact Information:
 - 1. Post a list of important telephone numbers.
 - a. Police and fire departments.
 - b. Ambulance service.
 - c. Contractor's home office.
 - d. Contractor's emergency after-hours telephone number.
 - e. Contractor's superintendent.
 - f. COR's office.
 - g. Engineers' offices.
 - h. Owner's office.
 - i. Principal subcontractors' field and home offices.

1.2 GENERAL SECURITY REGULATIONS

- A. Personnel List: Contractor shall provide the Resident Engineer with a list of Contractor's personnel who will require access to the site. The list shall be kept current during project work.
- B. Security Investigation: If contractors need access to active facility, Contractor's site superintendent shall submit to an FAA security background check and obtain an official FAA contractor ID badge. Other Contractor personnel may be subject to security investigation by FAA. Upon request by the Contracting Officer's Representative, the Contractor shall promptly complete all security forms provided by FAA.

- C. Communication: The Contractor shall request through the COR, a meeting with the Airport Manager and Control Tower personnel to discuss planned Contractor activities in the controlled airport operation area.
- D. Airport Requirements: Contractor must also meet all the Airport's security requirements for work on the AOA. FAA will not provide escorts, communication, or transportation.
- E. Nonpublicity: It is a specific condition of this Contract that the Contractor, or any subcontractors performing work on this project, shall not use or allow to be used any aspect of this project for publicity or advertising brochures without prior permission from the CO.
- F. FAA Security Regulations:
 - 1. All persons employed within the boundaries of the property or restricted-access areas therein, and all persons permitted to enter such property and areas shall comply with the security regulations that have been established for this Contract. See AMS 3.14-4 Access to FAA Facilities, Systems, Government Property, and Sensitive Unclassified Information (April 2024)
 - 2. The Contractor agrees on behalf of himself and all subcontractors that the security regulations will be observed by the Contractor and subcontractor personnel on the property. The Contractor shall make it a specific provision of his subcontracts that these regulations be accepted.
 - 3. At the commencement of the work under this Contract, coordinate security requirements with COR.
 - 4. Security Procedures: Personnel will require a security briefing that includes, as a minimum ALL of the following contract requirements:
 - a. Cameras, including cellular phones with internal cameras, are permitted on the Campus and are subject to use approval and review of photography by FAA when requested.
 - b. Personnel will not discuss the project or anything about the project with other than personnel who are assigned to the project and have a need-to-know.
 - c. Notify the COR when personnel are no longer employed by the Contractor or subcontractor.
 - d. Individuals identified as "Do Not Admit" will not be permitted access to the site.
 - 5. The COR reserves the right to close down the job site and order Contractor personnel off the premises in the event of an emergency or a shutdown, for as long as security problems persist. The Contractor may only return to the site with verbal approval from the COR and the CO or their designated representative.
 - 6. Personnel will be subject to inspection of their personal effects when entering and leaving the facility. In addition, unscheduled inspections of personnel may be made while on site.
 - 7. The COR reserves the right to exclude or remove from the site or building any employee of the Contractor or Subcontractor the COR deems incompetent, careless, insubordinate, or otherwise objectionable or whose continued employment on the work is deemed by the COR to be contrary to the public interests.

1.3 QUALITY ASSURANCE

- A. Provide high resolution digital photographs. Provide auxiliary lighting as required to produce clear, well-lit photographs without obscuring shadows.

- B. Depth of field is to be maximum range and within focus to clearly show Work. Photographs with blurry or out-of-focus areas will not be accepted.
- C. Clean Work prior to photographing, so that dirt and other foreign matter does not obscure detail.

1.4 COORDINATION

- A. Auxiliary Services: Cooperate with photographer and provide auxiliary services requested, including access to Project site and use of temporary facilities, including temporary lighting required to produce clear, well-lit photographs without obscuring shadows.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 47 00

SECTION 01 50 00

TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 DEFINITIONS

- A. Enclosure: As determined by COR, permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and all openings are closed with permanent construction or substantial temporary closures.

1.3 COOPERATE WITH OTHER ENTITIES

- A. Cooperate with the FAA and other public or private utility services, or a utility service of another government agency that may be authorized by the FAA to construct, reconstruct, or maintain such utility services or facilities during the progress of the work. Control operations to prevent the unscheduled interruption of such utility services and facilities.

1.4 NOTICE TO FAA/OPERATORS

- A. Prior to commencing the work in the general vicinity of an existing utility service or facility, Contractor must notify each FAA/operator in writing of activities that might affect its interests. If, in Contractor's opinion, the FAA/operator's assistance is needed to locate the utility service or facility or the presence of a representative of the FAA/operator is desirable to observe the work, such advice should be included in the notification. Furnish a copy of such written notices to COR

1.5 USE CHARGES

- A. General: Cost or use charges for temporary facilities shall be included in the Contract Sum. Allow other entities to use temporary services and facilities without cost, including, but not limited to, testing agencies, and authorities having jurisdiction.
- B. Sewer Service: Pay sewer service use charges for sewer usage by all entities for construction operations.
- C. Water Service: Pay water service use charges for water used by all entities for construction operations.
- D. Electric Power Service: Pay electric power service use charges for electricity used by all entities for construction operations.

1.6 SUBMITTALS

- A. Site Plan: Show temporary facilities, utility hookups, staging areas, and parking areas for construction personnel.

1.7 QUALITY ASSURANCE

- A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.
- B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.8 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility and cost for operation, maintenance, and protection of each permanent service during its use as a construction facility before FAA's acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Chain-Link Fencing: Minimum 2-inch, 0.148-inch- thick, galvanized steel, chain-link fabric fencing; minimum 8 feet high with galvanized steel pipe posts; minimum 2-3/8-inch- OD line posts and 2-7/8-inch- OD corner and pull posts with galvanized barbed-wire top strand.
- B. Lumber and Plywood: Comply with requirements in Section 06 10 00 ROUGH CARPENTRY.
- C. Gypsum Board: Minimum 1/2 inch thick by 48 inches wide by maximum available lengths; regular-type panels with tapered edges. Comply with ASTM C1396/C1396M REV Standard Specification for Gypsum Wallboard.
- D. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.
- E. Paint: Comply with requirements in Division 09 painting Sections.

2.2 TEMPORARY FACILITIES

- A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.
 - 1. Locate facilities to limit site disturbance.
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
 - 1. Arrange with utility company, FAA, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
- B. Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
 - 1. Connect temporary sewers to municipal system as directed by authorities having jurisdiction.
- C. Water Service: Install temporary water service and distribution piping in sizes and pressures adequate for construction.
- D. Sanitary Facilities: Provide temporary toilets, wash facilities, and drinking water for use of construction personnel. Comply with authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.
- E. Heating and cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.
- F. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.
- G. Electric Power Service: Provide temporary electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.
- H. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.
- I. Telephone Service: Provide FAA with a Contact number for the superintendent at each telephone, post a list of important telephone numbers.
 - a. Police and fire departments.
 - b. Ambulance service.
 - c. Contractor's home office.

- d. COR's office.
 - e. Engineers' offices.
 - f. Principal subcontractors' field and home offices.
2. Provide superintendent with cellular telephone.

- J. Internet Service: Provide internet connection for duration of project in common-use facilities and FAA field representative's office. Internet service will be of sufficient band width, speed etc. to efficiently view, download edit, etc. all BIM files, schedule files, and all contractor deliverables. The contractor shall supply internet service for FAA and FAA contractor employees working on site. For planning purposes, the contractor will plan for 10 connections to the internet all capable of performing required functions under this contract. Comply with Section 01 41 00 COORDINATION DRAWINGS AND BIM for BIM cloud-based computing requirements.

3.3 SUPPORT FACILITIES INSTALLATION

- A. General: Comply with the following:
- 1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 30 feet of building lines. Comply with NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations.
 - 2. Maintain support facilities until near Substantial Completion. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to FAA.
- B. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations.
- 1. Provide dust-control treatment that is nonpolluting and non-tracking. Reapply treatment as required to minimize dust.
- C. Traffic Controls: Comply with requirements of authorities having jurisdiction.
- 1. Protect existing site improvements to remain including curbs, pavement, and utilities.
 - 2. Maintain access for fire-fighting equipment and access to fire hydrants.
- D. Parking: Provide temporary parking areas for construction personnel.
- E. Dewatering Facilities and Drains: Comply with requirements of FAA and authorities having jurisdiction. Maintain Project site, excavations, and construction free of water.
- 1. Dispose of rainwater in a lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
 - 2. Remove snow and ice as required to minimize accumulations.
- F. Project Identification and Temporary Signs: Provide Project identification and other signs as indicated on Drawings and required by law. Install signs where indicated to inform public and individuals seeking entrance to Project. Unauthorized signs are not permitted.
- 1. Provide temporary, directional signs for construction personnel and visitors.
 - 2. Maintain and touchup signs so they are legible at all times.
- G. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of authorities having jurisdiction. ensure that waste-collection containers keep waste enclosed to prevent wind-blown obstructions on AOA

- H. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.
 - 1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.
- I. Temporary Elevator Use: Refer to Division 14 Sections for temporary use of new elevators.
- J. Temporary Stairs: Until permanent stairs are available, provide temporary stairs where ladders are not adequate.
- K. Temporary Use of Permanent Stairs: Cover finished, permanent stairs with protective covering of plywood or similar material so finishes will be undamaged at time of acceptance.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
 - 1. Comply with work restrictions specified in Section 01 10 00 SUMMARY OF WORK.
- B. Stormwater Control: Comply with authorities having jurisdiction. Provide barriers in and around excavations and subgrade construction to prevent flooding by runoff of stormwater from heavy rains.
- C. Pest Control: Engage pest-control service to recommend practices to minimize attraction and harboring of rodents, roaches, and other pests and to perform extermination and control procedures at regular intervals so Project will be free of pests and their residues at Substantial Completion. Obtain extended warranty for FAA. Perform control operations lawfully, using environmentally safe materials.
- D. Site Enclosure Fence: Before construction operations begin, furnish and install site enclosure fence in a manner that will prevent people and animals from easily entering site except by entrance gates.
 - 1. Extent of Fence: As indicated on Drawings.
 - 2. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide FAA with one set of keys.
- E. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.
- F. Barricades, Warning Signs, and Lights: Comply with requirements of authorities having jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.
- G. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.
 - 1. Where heating or cooling is needed and permanent enclosure is not complete, insulate temporary enclosures.

- H. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations .
1. Prohibit smoking in construction areas.
 2. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of authorities having jurisdiction.
 3. Develop and supervise an overall fire-prevention and -protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.5 DAMAGE TO SERVICES

- A. Should Contractor damage or interrupt the operation of a utility service or facility by accident or otherwise, it must immediately notify in writing the FAA/operator, appropriate public safety authorities, and the COR and must take all reasonable measures to prevent further damage or interruption of service. Cooperate with the utility service or facility FAA and the COR continuously until such damage has been repaired and service restored

3.6 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.
 2. The Contractor and each subcontractor shall be responsible for cleaning and maintaining all temporary offices and storage sheds in proper condition acceptable to the COR. All exposed surfaces on the outside and inside of field offices and temporary toilet enclosures and outside of storage sheds shall be painted and maintained with exterior enamel paint. Colors are subject to approval by the COR. All temporary facilities shall be maintained by the Contractor and shall be kept in usable condition at all times until completion of the work and/or their removal is authorized by the COR.
 3. Maintain lighting. Promptly replace worn or defective parts and non-working bulbs.
 4. Maintain temporary water system: Maintain system to provide continuous service with adequate pressure to outlets. Maintain connections, pipes, fittings, and fixtures and conserve use of all utilities. Failure to stop leaks or other waste of water will be cause for revocation of permit for the use of said water from the airport system.
 5. Maintain temporary toilet facilities: Clean facilities and surrounding areas daily. Provide toilet paper, paper towels and soap in suitable dispensers.
 6. Maintain erosion, sedimentation, and pollution control measures throughout the project life cycle. Inspect erosion, sedimentation, and pollution control measures once per week and following each rain event., Any deficiencies shall be reported to the COR and promptly corrected within 48-hours.

- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
 - 1. Materials and facilities that constitute temporary facilities are property of Contractor. FAA reserves right to take possession of Project identification signs.
 - 2. Consider inserting specific removal requirements, as illustrated in first subparagraph below.
 - 3. Remove temporary paving not intended for or acceptable for integration into permanent paving. Where area is intended for landscape development, remove soil and aggregate fill that do not comply with requirements for fill or subsoil. Remove materials contaminated with road oil, asphalt and other petrochemical compounds, and other substances that might impair growth of plant materials or lawns. Repair or replace street paving, curbs, and sidewalks at temporary entrances, as required by authorities having jurisdiction.
 - 4. Remove temporary erosion, sedimentation and pollution control measures upon final stabilization of site.
 - 5. Remove temporary lighting material and equipment when permanent system is operational.
 - 6. Remove temporary toilet facilities when permanent facilities are available for use, but no later than Substantial Completion.
 - 7. At Substantial Completion, clean and renovate permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 01 77 00 PROJECT CLOSEOUT.

END OF SECTION 01 50 00

SECTION 01 55 26

MAINTENANCE OF TRAFFIC

PART 1 - GENERAL

1.1 DESCRIPTION

- A. The work to be performed under this Section shall consist of all work and all measures to be employed to maintain the efficient and orderly movement of traffic in the area of construction as shown on the Plans and as described in this Section.

1.2 MANAGEMENT OF TRAFFIC

- A. The Contractor shall provide, install, and maintain the temporary traffic control devices, furnish flaggers, and perform all work required to conform to the provisions of this Section.
- B. The Contract Documents show the general location of signs, lights, markings, delineators, special lighting, guardrails, barricades, temporary pavements, flagger stations, and other temporary devices and work required to control traffic at each work sequence area. These and any other measures shall be provided by the contractor to ensure proper traffic control.
- C. Before commencing work in any area, the Contractor shall install the temporary traffic control devices, stations, etc., at the work site, and he shall obtain the approval of the COR before commencing any work that affects, in any way, the existing traffic flow.
- D. At least one lane of traffic shall be maintained at all times on Dooley Road.

PART 2 - PRODUCTS

2.1 TRAFFIC CONTROL DEVICES, WARNING DEVICES AND BARRIERS

- A. Traffic control devices, warning devices and barriers shall meet the applicable requirements of the current edition of the Department of Transportation Standard Specifications for Road and Bridge Construction and the FHWA Manual or Uniform Traffic Control Devices (MUTCD); subject to COR's approval.

PART 3 - EXECUTION

3.1 MAINTENANCE OF TEMPORARY TRAFFIC CONTROL DEVICES, PAVEMENTS, AND FACILITIES

- A. The Contractor shall maintain all traffic control devices in proper repair and working order. The Contractor shall also maintain all pavements constructed or utilized for temporary traffic movement, and shall maintain all other traffic service facilities such as guardrail, area lighting, etc., necessary for the efficient and orderly movement of traffic within the construction area.

- B. In the event of the Contractor's failure to properly maintain any of these devices, pavements or facilities, the FAA may cause such maintenance, as it deems necessary, to be performed by its own or another Contractor's forces and the costs of such maintenance shall be deducted from monies due the Contractor for work performed under this Contract.

3.2 INTERFERENCE WITH TRAFFIC

- A. The Contractor shall conduct his work so as to cause no unnecessary interference with traffic and it shall comply with all requirements governing its employee parking, areas prohibited to his operation, and access routes to authorized work areas.
- B. The Contractor shall not permit its workers and equipment to interfere with the movement of traffic in those areas adjacent to its work areas. The Contractor shall not obstruct sight lines, create obstructions to lighting nor create hazards or nuisance by allowing spills or wind transported materials to accumulate in traffic areas.
- C. Traffic control personnel shall be provided at all times that any travel lane is obstructed. These personnel shall be stationed at each end of the obstructed area and shall provide safe passage of vehicles through the obstructed area.
- D. All lanes of travel shall be unobstructed at night and at all times when traffic directors are not present. Metal plates shall be provided and secured in place if pavement is not provided.
- E. The Contractor shall promptly remove any spills or wind transported debris occurring on traveled roadways.

3.3 FINAL CLEANUP

- A. After work has been completed, the Contractor shall remove all temporary traffic control devices, temporary pavements and other temporary work and devices installed for traffic control. The Contractor shall restore the site to its original condition or to the revised condition shown on the Plans.

END OF SECTION 01 55 26

SECTION 01 65 00

PRODUCT DELIVERY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Description of Work:
 - 1. General requirements for delivery, inspection, storage and handling of materials and equipment.

1.2 RELATED WORK SPECIFIED ELSEWHERE

- A. See specific technical Specification Sections for additional requirements relating to delivery, storage, handling, installation and field quality control.

1.3 DELIVERY, STORAGE AND HANDLING

- A. Packaging and Transportation:
 - 1. Require supplier to package finished products in boxes or crates for protection during shipment, handling and storage. Protect sensitive products against exposure to elements and moisture.
 - 2. Protect sensitive equipment and finishes against impact, abrasion and other damage.
- B. Delivery and Receiving:
 - 1. Arrange deliveries of products in accordance with construction schedule. Allow time for inspection prior to installation.
 - 2. Coordinate deliveries to avoid conflict with Work and conditions at site, limitations on storage space and availability of personnel and handling equipment.
 - 3. Deliver products in undamaged, dry condition, in original unopened containers or packaging with identifying labels intact and legible.
 - 4. Clearly mark partial deliveries of component parts of equipment to identify equipment and contents to permit easy accumulation of parts and to facilitate assembly.
 - 5. Immediately on delivery, inspect shipment to assure:
 - a. Product complies with requirements of Contract Documents and reviewed submittals.
 - b. Quantities are correct.
 - c. Accessories and installation hardware are correct.
 - d. Containers and packages are intact and labels legible.
 - e. Products are protected and undamaged.
- C. Product Handling:
 - 1. Provide equipment and personnel to handle products by methods to prevent soiling and damage.
 - 2. Provide additional protection during handling to prevent marring and otherwise damaging products, packaging and surrounding surfaces.

3. Handle product by methods to avoid bending or overstressing. Lift large and heavy components only at designated lift points.

D. Storage:

1. Store products, immediately on delivery, in accordance with manufacturer's instructions, with seals and labels intact. Protect until installed.
2. Arrange storage to provide access for maintenance of stored items and for inspection.
3. Enclosed storage:
 - a. Store products subject to damage by elements in substantial weathertight enclosures.
 - b. Maintain temperature and humidity within ranges stated in manufacturer's instructions.
 - c. Provide humidity control and ventilation for sensitive products as required by manufacturer's instructions.
 - d. Store unpacked and loose products on shelves, in bins or in neat groups of like items.
4. Exterior storage:
 - a. Provide substantial platforms, blocking or skids to support fabricated products above ground; slope to provide drainage.
 - b. Protect products from soiling and staining.
 - c. For products subject to discoloration or deterioration from exposure to elements, cover with impervious sheet material.
 - d. Provide ventilation to avoid condensation.
 - e. Store loose granular materials on clean, solid surfaces such as rigid sheet materials or pavement. Prevent mixing with foreign matter.
 - f. Prevent mixing of refuse or chemically injurious materials or liquids with building materials.
5. Periodically inspect stored products to verify proper storage.

PART 2 - PRODUCTS

- A. Not Used

PART 3 - EXECUTION

- A. Not Used

END OF SECTION 01 65 00

SECTION 01 71 33

PROTECTION OF WORK AND PROPERTY

PART 1 - GENERAL

1.1 SUMMARY

- A. Protection of products after installation.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

- A. Protect installed products and control traffic in immediate area to prevent damage from subsequent operations.
- B. Provide protective coverings at walls, projections, corners and jambs, sills and soffits of openings in and adjacent to traffic areas.
- C. Cover walls and floors of elevator cabs and jambs of cab doors with 3/4 inch plywood, when elevators are used by construction personnel.
- D. Protect finished floors and stairs from dirt, wear and damage:
 - 1. Secure heavy sheet goods or similar protective materials in place, in areas subject to foot traffic.
 - 2. Lay planking or similar rigid materials in place, in areas subject to movement of heavy objects.
 - 3. Lay planking or similar rigid materials in place, in areas where storage of products will occur.
- E. Protect waterproofed and roofed surfaces:
 - 1. Restrict use of surfaces from traffic of any kind and from storage of products.
 - 2. When an activity is mandatory, obtain recommendations for protection of surfaces from manufacturer. Install protection and remove on completion of activity. Restrict use of adjacent unprotected areas.
- F. Restrict traffic of any kind across planted lawn and landscape areas.

END OF SECTION 01 71 33

SECTION 01 71 43
CONSTRUCTION CLEANING

PART 1 - GENERAL

1.1 REQUIREMENT INCLUDED

- A. Cleaning and disposal of waste materials, debris and rubbish during construction.

PART 2 - PRODUCTS

2.1 EQUIPMENT

- A. Provide covered containers for deposit of waste materials, debris and rubbish.

PART 3 - EXECUTION

3.1 CLEANING

- A. Maintain areas under Contractor's control free of waste materials, scraps, surplus material, debris and rubbish. Maintain site in a clean and orderly condition.
- B. Remove debris and rubbish from pipe chases, plenums attics, crawl spaces and other closed or remote spaces, prior to closing the space.
- C. Clean interior areas daily to provide suitable conditions for work and to prevent fire or accidents.
- D. Use power brooms to clean paved areas as needed and immediately prior to opening any paved area to vehicular traffic.
- E. All combustible waste materials shall be removed from buildings at the end of each working day.
- F. Broom clean interior areas prior to start of surface finishing and continue cleaning on a daily basis.
- G. Control cleaning operations so that dust and other particulates will not adhere to wet or newly-coated surfaces.
- H. Responsibility for construction cleaning shall not be delegated to subcontractors performing construction work under this Contract.

3.2 DISPOSAL

- A. Remove waste materials, debris and rubbish from site bi-weekly and legally dispose of off-site in an authorized disposal area.

3.3 CONTRACTOR'S FAILURE TO CLEAN

- A. If the Contractor fails to maintain levels of cleanliness in work areas, satisfactory to the COR, then the FAA shall have the right to cause such areas to be cleaned by others. The costs to the FAA for such cleaning, plus 25% for administration, shall be the obligation of the Contractor and shall be deducted from any money due the Contractor hereunder.

END OF SECTION 01 71 43

SECTION 01 73 19

INSTALLATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Description of Work: General requirements for installation and protections of materials and equipment.

1.2 RELATED WORK SPECIFIED ELSEWHERE

- A. See specific technical Specification Sections for additional requirements relating to installation, protection and quality control.

PART 2 - PRODUCTS

- A. Not Used

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General:
 - 1. Before product installation, verify space limitations, clearances, accessibility and other Project conditions that may affect installation. Notify COR if conditions are not satisfactory.
 - 2. Start of Work constitutes acceptance of conditions by Contractor.
- B. Manufacturer's Instructions:
 - 1. Manufactured articles, materials and equipment shall be applied, installed, connected, erected, used, cleaned and conditioned in accordance with the manufacturer's written or printed directions and instructions unless otherwise indicated in the Contract Documents.
 - 2. Maintain one set of installation instructions at job site during installation and until completion.
 - 3. Comply with instructions in full detail, including each step in sequence.
 - 4. If job conditions or specified requirements conflict with manufacturer's instructions, consult COR before proceeding.
- C. Workmanship:
 - 1. Comply with industry standards specified except when more restrictive tolerances or specified requirements indicate more rigid standards or more precise workmanship.
 - 2. Where no explicit quality or standards for materials or workmanship are established for Work, such Work shall be of good quality for the intended use and consistent with the quality of the surrounding Work and of the construction of the Project generally.

- D. Anchorage: Secure products with positive anchorage devices designed and sized to withstand stresses, vibration and racking.
- E. Mounting Heights:
 - 1. Where mounting heights are not indicated, mount individual units of Work at industry-recognized standard mounting heights for applications indicated.
 - 2. Refer questionable mounting height choices to COR before proceeding.
 - 3. Obtain exact locations of fixtures and outlets from COR before Work is roughed in; Work installed without such information from COR shall be relocated at Contractor's expense.

3.2 STARTING AND ADJUSTING

- A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.
- B. Adjust equipment for proper operation. Adjust operating components for proper operation without binding.
- C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Lubricate moving parts. Replace damaged and malfunctioning controls and equipment.

3.3 FIELD QUALITY CONTROL

- A. Manufacturer's Field Services: When required by Contract Documents, have manufacturer provide qualified representative to observe field conditions, conditions of surfaces and installation, quality of workmanship; to start-up equipment and to test, adjust and balance equipment as applicable.
- B. Contractor's Quality Control: Maintain quality control over supervision, Subcontractors, suppliers, manufacturers, products, services, workmanship and site conditions, to produce Work in accordance with requirements of Contract Documents.

3.4 PROTECTION

- A. Protect installed materials from damage until Substantial Completion. Repair or replace materials damaged during construction operations.

END OF SECTION 01 73 19

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes administrative and procedural requirements for the following:
 - 1. Recycling nonhazardous construction waste.
 - 2. Disposing of nonhazardous demolition and construction waste.

1.3 DEFINITIONS

- A. Construction Waste: Building, structure, and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Site improvement materials resulting from clearing operations.
- C. Disposal: Removal of demolition or construction waste and subsequent salvage, sale, recycling, or deposit in landfill, incinerator acceptable to authorities having jurisdiction, or designated spoil areas on Owner's property.
- D. Recycle: Recovery of construction waste for subsequent processing in preparation for reuse.
- E. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.
- F. Salvage and Reuse: Recovery of demolition or construction waste and subsequent incorporation into the Work.

1.4 MATERIALS OWNERSHIP

- A. Unless otherwise indicated, demolition and construction waste becomes property of Contractor.

1.5 ACTION SUBMITTALS

- A. Waste Management Plan: Submit plan within 7 days of date established for the Notice to Proceed.

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with transportation and disposal regulations of authorities having jurisdiction.
- B. Waste Management Conference(s): Conduct conference(s) at Project site to comply with requirements in Section 013100 "Project Management and Coordination." Review methods and procedures related to waste management including, but not limited to, the following:
 - 1. Review and discuss waste management plan including responsibilities of each contractor.
 - 2. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
 - 3. Review waste management requirements for each trade.

1.7 WASTE MANAGEMENT PLAN

- A. General: Develop a waste management plan according to requirements in this Section or adhere to the Construction Waste Management Plan. Plan shall consist of waste identification. Indicate quantities by weight or volume, but use same units of measure throughout waste management plan.
- B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing, and construction waste generated by the Work. Use Form CWM-1 for construction waste or the Contractor's form designation for a combined waste. Include estimated quantities and assumptions for estimates.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

- A. General: Implement approved waste management plan. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during the entire duration of the Contract.
 - 1. Comply with operation, termination, and removal requirements in Section 015000 "Temporary Facilities and Controls."
- B. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work.
 - 1. Distribute waste management plan to everyone concerned within three days of submittal return.
 - 2. Distribute waste management plan to entities when they first begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.

- C. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
 - 1. Designate and label specific areas on Project site necessary for separating materials that are to be salvaged and recycled.
 - 2. Comply with Section 015000 "Temporary Facilities and Controls" for controlling dust and dirt, environmental protection, and noise control.

3.2 DISPOSAL OF WASTE

- A. General: Except for items or materials to be salvaged or recycled, remove waste materials from Project site and legally dispose of them in a landfill or incinerator acceptable to authorities having jurisdiction.
 - 1. Except as otherwise specified, do not allow waste materials that are to be disposed of accumulate on-site.
 - 2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
- B. General: Except for items or materials to be salvaged or recycled, remove waste materials and legally dispose of at designated spoil areas on Owner's property.
- C. Burning: Do not burn waste materials.

END OF SECTION 01 74 19

SECTION 01 77 00

PROJECT CLOSEOUT

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:
1. Compliance with Specification 01 77 10-FINAL CLEANING
 2. Compliance with Specification 01 78 23-OPERATION AND MAINTENANCE DATA
 3. Compliance with Specification 01 78 26-WARRANTIES AND GUARANTEES
 4. Compliance with Specification 01 78 29-PROJECT RECORD DOCUMENTS
 5. Completion of Asbestos and Lead Free Certification
 6. Final Punch List

1.2 SUBSTANTIAL COMPLETION

- A. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion, complete the following. List items below that are incomplete in request.
1. Prepare a list of items to be completed and corrected (punch list), the value of items on the list, and reasons why the Work is not complete.
 2. Advise FAA of pending insurance changeover requirements.
 3. Obtain and submit releases permitting FAA unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
 4. Prepare and submit Coordination Drawings, Project Record Documents, Facility Data operation and maintenance manuals, Final Completion construction photographs, damage or settlement surveys, property surveys, and similar final record information.
 5. Deliver tools, spare parts, extra materials, and similar items to location designated by FAA. Label with manufacturer's name and model number where applicable.
 6. Make final changeover of permanent locks and deliver keys to FAA. Advise FAA's personnel of changeover in security provisions.
 7. Complete startup testing of systems.
 8. Provide a final TAB report and preliminary commissioning report. The preliminary commissioning report shall include all documents required from the beginning of the contract through the verification of all control sequences and mechanical equipment operations. There shall be no items that are reported in less than complete working order per the contract documents.
 9. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.
 10. Advise FAA of changeover in utilities.
 11. Submit changeover information related to FAA's occupancy, use, operation, and maintenance.
 12. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.

- B. Inspection: Submit a written request for inspection for Substantial Completion, also referred to as the Contractor Acceptance Inspection (CAI). On receipt of request, COR will either schedule the inspection within 14 days or notify Contractor of unfulfilled requirements. COR will prepare the Certificate of Substantial Completion after the inspection or will notify Contractor of items, either on Contractor's list or additional items identified by COR, that must be completed or corrected before certificate will be issued. COR will also provide a punch list that will form the basis of requirements for the Final Completion.

1.3 FINAL COMPLETION

- A. Preliminary Procedures: Contractor should request final inspection prior to contract completion date. Before requesting final inspection for determining date of Final Completion, complete the following:
 - 1. Submit a Final Application for Payment.
 - 2. Submit certified copy of COR's Substantial Completion inspection list (punch list) of items to be completed or corrected, endorsed and dated by COR. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
 - 3. Submit specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
 - 4. Submit a letter from the airport certifying that work areas located on the airport were left in a satisfactory condition.
 - 5. Perform a final cleaning in accordance with Section 01 77 10 FINAL CLEANING.
- B. Inspection: Submit a written request for final inspection for acceptance. On receipt of request, COR will either proceed with inspection or notify Contractor of unfulfilled requirements. COR will prepare a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.

1.4 WARRANTIES

- A. Submit warranties in accordance with Section 01 78 36 WARRANTIES AND GUARANTEES. Warranty period shall begin on date of Substantial Completion as listed in Certificate of Substantial Completion.
- B. Partial Occupancy: Submit properly executed warranties within fifteen (15) days of completion of designated portions of the Work that are completed and occupied or used by FAA during construction period by separate agreement with Contractor.
- C. Provide additional copies of each warranty to include in operation and maintenance manuals.

1.5 LOCATION CERTIFICATION

- A. Complete and submit the location accuracy certification form found immediately following this section to the COR.

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 77 00

SECTION 01 77 10

FINAL CLEANING

PART 1 - GENERAL

1.1 REQUIREMENT INCLUDED

- A. Final cleaning of project.

1.2 SUMMARY

- A. This section includes administrative and procedural requirements, protections of construction in progress, and for final cleaning at Substantial Completion.
- B. Environmental Requirements: Conduct cleaning and waste-disposal operations in compliance with local laws and ordinances. Comply fully with federal and local environmental and antipollution regulations.
- C. Do not dispose of volatile wastes, such as mineral spirits, oil, or paint thinner, in storm or sanitary drains.
- D. Burning or burying of debris, rubbish, or other waste material on the premises is not permitted.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Cleaning Agents: Use cleaning materials and agents recommended by the manufacturer or fabricator of the material to be cleaned. Do not use cleaning agents that are potentially hazardous to health, property or that might damage finished surfaces.

PART 3 - EXECUTION

3.1 FINAL CLEANING

- A. General: Provide final-cleaning operations. Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit of work to the condition expected from a commercial building cleaning and maintenance program. Comply with manufacturer's instructions.
- B. Cleaning Operations: Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for the entire Project or a portion of the Project.
 - 1. Clean the Project Site, yard and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and foreign substances.

2. Sweep paved areas broom clean. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.
 3. Broom and mop clean concrete floors in unoccupied spaces.
 4. Remove petrochemical spills, stains, and other foreign deposits.
 5. Remove tools, construction equipment, machinery, and surplus material from the site.
 6. Vacuum clean carpet and similar soft surfaces, removing debris and excess nap. Shampoo carpet as directed by COR.
 7. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
 8. Remove marks, stains, fingerprints, and other soils or other dirt from painted, decorated, and natural finished woodwork and other work.
 9. Clean cabinet work removing stains, paint, dirt and dust.
 10. Remove spots, plaster, soil and paint from ceramic tile, marble, and other finished materials, and wash or wipe clean.
 11. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other substances that are noticeable vision-obscuring materials. Replace chipped or broken glass and other damaged transparent materials. Polish mirrors and glass, taking care not to scratch surfaces.
 12. Clean flooring materials thoroughly, including stripping, buffing and waxing. Comply with materials manufacturer's instructions and recommendations.
 13. Remove labels that are not permanent labels.
 14. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
 15. Clean food-service equipment to a sanitary condition, ready and acceptable for its intended use.
 16. Wipe surfaces of mechanical and electrical equipment, elevator equipment, and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
 17. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.
 18. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out bulbs and defective and noisy starters.
 19. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
 20. Clean ductwork, blowers, and coils of units that were operated during construction.
 21. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
 22. Leave the Project clean and ready for occupancy.
- C. Pest Control: Engage an experienced, licensed exterminator to make a final inspection and rid the Project of rodents, insects and other pests. Comply with regulations of local authorities.
- D. Removal of Protection: Remove temporary protection and facilities installed during construction to protect previously completed installations during the remainder of the construction period.
1. Where extra materials of value remain after completion of associated work, they become the FAA's property. Dispose of these materials as directed by the FAA.

2. The Contractor shall not dispose of debris or waste materials on the FAA's property without the prior written approval of the FAA.

END OF SECTION 01 77 10

OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including the following:
 - 1. Operation and maintenance documentation directory.
 - 2. Emergency manuals.
 - 3. Operation manuals for systems, subsystems, and equipment.
 - 4. Maintenance manuals for the care and maintenance of products, materials, and finishes; systems and equipment.
 - 5. Manual for sustainable operations.

1.2 DEFINITIONS

- A. System: An organized collection of parts, equipment, or subsystems united by regular interaction.
- B. Subsystem: A portion of a system with characteristics similar to a system.

1.3 SUBMITTALS

- A. Initial Submittal: Submit 2 bound draft copies of each manual and electronic copies in PDF format at least 15 days before requesting inspection for Substantial Completion. Include a complete operation and maintenance directory. COR will return one copy of draft and mark whether general scope and content of manual are acceptable.
- B. Final Submittal: Submit two bound copies of each manual and two electronic copies (PDFs) in final form at least 15 days before final inspection. If modifications are required, COR will return one copy with comments within 15 days after final inspection.
 - 1. Correct or modify each manual to comply with COR's comments. Submit two (2) bound copies and two (2) electronic copies of each corrected manual within 15 days of receipt of COR's comments.

1.4 COORDINATION

- A. Where operation and maintenance documentation includes information on installations by more than one factory-authorized service representative, assemble and coordinate information furnished by representatives and prepare manuals.

PART 2 - PRODUCTS

2.1 OPERATION AND MAINTENANCE DOCUMENTATION DIRECTORY

- A. Organization: Include a section in the directory for each of the following:
 - 1. List of documents.

2. List of systems.
 3. List of equipment.
 4. Table of contents.
- B. List of Systems and Subsystems: List systems alphabetically. Include references to operation and maintenance manuals that contain information about each system.
- C. List of Equipment: List equipment for each system, organized alphabetically by system. For pieces of equipment not part of system, list alphabetically in separate list.
- D. Tables of Contents: Include a table of contents for each emergency, operation, and maintenance manual.
- E. Identification: In the documentation directory and in each operation and maintenance manual, identify each system, subsystem, and piece of equipment with same designation used in the Contract Documents. If no designation exists, assign a designation according to ASHRAE Guideline 4, "Preparation of Operating and Maintenance Documentation for Building Systems."

2.2 MANUALS (GENERAL)

- A. Organization: Unless otherwise indicated, organize each manual into a separate section for each system and subsystem, and a separate section for each piece of equipment not part of a system. Each manual shall contain the following materials, in the order listed:
1. Title page.
 2. Table of contents.
 3. Manual contents.
- B. Title Page: Enclose title page in transparent plastic sleeve. Include the following information:
1. Subject matter included in manual.
 2. Name and address of Project.
 3. Address of FAA.
 4. Date of submittal.
 5. Name, address, and telephone number of Contractor.
 6. Cross-reference to related systems in other operation and maintenance manuals.
- C. Table of Contents: List each product included in manual, identified by product name, indexed to the content of the volume, and cross-referenced to Specification Section number in Project Manual.
1. If operation or maintenance documentation requires more than one volume to accommodate data, include comprehensive table of contents for all volumes in each volume of the set.
- D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. If possible, assemble instructions for subsystems, equipment, and components of one system into a single binder.
1. Binders: Heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2-by-11-inch paper; with clear plastic sleeve on spine to hold label describing contents and with pockets inside covers to hold folded oversize sheets.

- a. If two or more binders are necessary to accommodate data of a system, organize data in each binder into groupings by subsystem and related components. Cross-reference other binders if necessary to provide essential information for proper operation or maintenance of equipment or system.
 - b. Identify each binder on front and spine, with printed title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.
2. Dividers: Heavy-paper dividers with plastic-covered tabs for each section. Mark each tab to indicate contents. Include typed list of products and major components of equipment included in the section on each divider, cross-referenced to Specification Section number and title of Project Manual.
3. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
4. Supplementary Text: Prepared on 8-1/2-by-11-inch white bond paper.
5. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.
 - a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
 - b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.

2.3 EMERGENCY MANUALS

- A. Content: Organize manual into a separate section for each of the following:
 1. Type of emergency.
 2. Emergency instructions.
 3. Emergency procedures.
- B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component:
 1. Fire.
 2. Flood.
 3. Gas leak.
 4. Water leak.
 5. Power failure.
 6. Water outage.
 7. System, subsystem, or equipment failure.
 8. Chemical release or spill.
- C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of FAA's operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.
- D. Emergency Procedures: Include the following, as applicable:
 1. Instructions on stopping.
 2. Shutdown instructions for each type of emergency.
 3. Operating instructions for conditions outside normal operating limits.
 4. Required sequences for electric or electronic systems.

5. Special operating instructions and procedures.

2.4 OPERATION MANUALS

- A. Content: In addition to requirements in this Section, include operation data required in individual Specification Sections and the following information:
 1. System, subsystem, and equipment descriptions.
 2. Performance and design criteria if Contractor is delegated design responsibility.
 3. Operating standards.
 4. Operating procedures.
 5. Operating logs.
 6. Wiring diagrams.
 7. Control diagrams.
 8. Piped system diagrams.
 9. Precautions against improper use.
 10. License requirements including inspection and renewal dates.
- B. Descriptions: Include the following:
 1. Product name and model number.
 2. Manufacturer's name.
 3. Equipment identification with serial number of each component.
 4. Equipment function.
 5. Operating characteristics.
 6. Limiting conditions.
 7. Performance curves.
 8. Engineering data and tests.
 9. Complete nomenclature and number of replacement parts.
- C. Operating Procedures: Include the following, as applicable:
 1. Startup procedures.
 2. Equipment or system break-in procedures.
 3. Routine and normal operating instructions.
 4. Regulation and control procedures.
 5. Instructions on stopping.
 6. Normal shutdown instructions.
 7. Seasonal and weekend operating instructions.
 8. Required sequences for electric or electronic systems.
 9. Special operating instructions and procedures.
- D. Systems and Equipment Controls: Describe the sequence of operation, and diagram controls as installed.
- E. Piped Systems: Diagram piping as installed, and identify color-coding where required for identification.

2.5 PRODUCT MAINTENANCE MANUAL

- A. Content: Organize manual into a separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

- B. Source Information: List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
- C. Product Information: Include the following, as applicable:
 - 1. Product name and model number.
 - 2. Manufacturer's name.
 - 3. Color, pattern, and texture.
 - 4. Material and chemical composition.
 - 5. Reordering information for specially manufactured products.
- D. Maintenance Procedures: Include manufacturer's written recommendations and the following:
 - 1. Inspection procedures.
 - 2. Types of cleaning agents to be used and methods of cleaning.
 - 3. List of cleaning agents and methods of cleaning detrimental to product.
 - 4. Schedule for routine cleaning and maintenance.
 - 5. Repair instructions.
- E. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.
- F. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
 - 1. Include procedures to follow and required notifications for warranty claims.

2.6 SYSTEMS AND EQUIPMENT MAINTENANCE MANUAL

- A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.
- B. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
- C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:
 - 1. Standard printed maintenance instructions and bulletins.
 - 2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
 - 3. Identification and nomenclature of parts and components.
 - 4. List of items recommended to be stocked as spare parts.
- D. Maintenance Procedures: Include the following information and items that detail essential maintenance procedures:
 - 1. Test and inspection instructions.
 - 2. Troubleshooting guide.

3. Precautions against improper maintenance.
 4. Disassembly; component removal, repair, and replacement; and reassembly instructions.
 5. Aligning, adjusting, and checking instructions.
 6. Demonstration and training videotape, if available.
- E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
 2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.
- F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
- G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.
- H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
1. Include procedures to follow and required notifications for warranty claims.
- 2.7 MANUAL FOR SUSTAINABLE OPERATIONS
- A. Provide all information required to operate building and its systems and components in an environmentally sustainable manner.
1. Arrange manual to be used for training building staff.
 2. Include section on Integrated Pest Management (IPM).

PART 3 - EXECUTION

3.1 MANUAL PREPARATION

- A. Operation and Maintenance Documentation Directory: Prepare a separate manual that provides an organized reference to emergency, operation, and maintenance manuals.
- B. Emergency Manual: Assemble a complete set of emergency information indicating procedures for use by emergency personnel and by FAA's operating personnel for types of emergencies indicated.
- C. Product Maintenance Manual: Assemble a complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into the Work.
- D. Operation and Maintenance Manuals: Assemble a complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
 2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by FAA's operating personnel.
- E. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark each sheet to identify each product or component incorporated into the Work. If data include more than one item in a tabular format, identify each item using appropriate references from the Contract Documents. Identify data applicable to the Work and delete references to information not applicable.
1. Prepare supplementary text if manufacturers' standard printed data are not available and where the information is necessary for proper operation and maintenance of equipment or systems.
- F. Drawings: Prepare drawings supplementing manufacturers' printed data to illustrate the relationship of component parts of equipment and systems and to illustrate control sequence and flow diagrams. Coordinate these drawings with information contained in Record Drawings to ensure correct illustration of completed installation.
1. Do not use original Project Record Documents as part of operation and maintenance manuals.
 2. Comply with requirements of newly prepared Record Drawings.
- G. Comply with Section 01 77 00 PROJECT CLOSEOUT for schedule for submitting operation and maintenance documentation.

END OF SECTION 01 78 23

SECTION 01 78 26

WARRANTIES AND GUARANTEES

PART 1 - GENERAL

1.1 REQUIREMENTS INCLUDED

- A. Preparation and submittal of warranties and guarantees.

1.2 INFORMATION SUBMITTALS

- A. Form of Warranty:
 - 1. Bind in commercial quality 8 ½ x 11 inch three-ring side binders, with hardback, cleanable, plastic covers.
 - 2. Label cover of each binder with typed or printed title 'WARRANTIES AND GUARANTEES', with Contract No. and Project Title; name, address and telephone number of Contractor.
 - 3. Table of Contents: Neatly typed, in the sequence of the Table of Contents of the Project Manual, with each item identified with the number and title of the specification section in which specified and the name of the product or work item.
 - 4. Separate each warranty or guaranty with index tab sheets keyed to the Table of Contents listing. Provide full information, using separate typed sheet as necessary. List subcontractor, supplier and manufacturer, with name, address and telephone number of responsible principal.

1.3 PREPARATION OF WARRANTY

- A. Obtain warranties and guarantees, executed in duplicate by responsible subcontractors, suppliers and manufacturers, within ten (10) days after completion of the applicable item of work. Date of beginning of time of warranty will be the date of Substantial Completion.
- B. Warranties and guarantees shall be made out in the name of, and accrue to the benefit of the Federal Aviation Administration.

1.4 TIME OF WARRANTY

- A. Provide warranties prior to final acceptance.
- B. For items of work when acceptance is delayed beyond date of Substantial Completion, submit within ten (10) days after acceptance, listing the date of acceptance as the beginning of the warranty or guaranty period.

1.5 EQUIPMENT WARRANTY TAGS AND GUARANTEE LOCAL REPRESENTATIVES

- A. The Contractor shall furnish with each guarantee, the name address, and telephone number of the guarantor, the name, address, and telephone number of the guarantor's representative nearest to the site, who, upon request of the FAA representative, will honor the guarantee during the guaranty

period and will provide the service prescribed by the terms of the guarantee. At the time of installation, the Contractor shall tag each item of warranted equipment with a durable, oil and water resistant tag approved by the Contracting Officer's Representative (COR). Tag shall be attached with copper wire and sprayed with a clear silicone, waterproof coating. Leave the date of acceptance and inspectors signature blank until project is accepted for Substantial Completion. Tag shall show the following information:

- B. Equipment warranty tags
 - 1. Type of Equipment
 - 2. Accepted Date
 - 3. Warranted Until
 - 4. Under Contract Number
 - 5. Inspector's Signature

1.6 QUANTITY

- A. Provide one (1) complete digital copy and two (2) complete physical copies of warranties and guarantees.
- B. Tabs by spec section number. TOC

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 78 26

SECTION 01 78 29

PROJECT RECORD DOCUMENTS

PART 1 - GENERAL

1.1 REQUIREMENTS INCLUDED

- A. This section describes the requirements for the creation and maintenance of “As Built Drawings;” referred to herein as Record Documents.
- B. Maintenance of Record Documents.
- C. Submittal of Record Documents.

1.2 MAINTENANCE OF DOCUMENTS AND SAMPLES

- A. Store Record Documents in Field Office apart from documents used for construction. Provide files, racks and secure storage for Record Documents.
- B. Maintain Record Documents in clean, dry and legible conditions. Do not use Record Documents for construction purposes.
- C. Keep Record Documents and Samples available for inspection by FAA.

1.3 RECORD DOCUMENTS (AS-BUILT) INFORMATION

- A. Record information on a set of full size drawings. Provide redlines on paper and digitally.
- B. Provide felt tip marking pens, maintaining separate colors for each major system, for recording information.
- C. Record information concurrently with construction progress. Do not conceal any work until required information is recorded.
- D. Contract Drawings and approved Shop Drawings: Legibly mark each item to record actual construction, including:
 - 1. Measured depths of elements of foundation in relation to finish grade or first floor datum.
 - 2. Measured horizontal and vertical locations of underground utilities and appurtenances, referenced to permanent surface improvements.
 - 3. Measured locations of internal utilities and appurtenances concealed in construction, referenced to visible and accessible features of construction.
 - 4. Field changes of dimensions and details.
 - 5. Changes made by Addenda, Change Order(s) (if any) and Work Order(s) (if any).
 - 6. Details not on original Contract Drawings.
 - 7. References to related Shop Drawings and Modifications.

- E. Specifications: Legibly mark each item to record actual construction, including changes made by Addenda and Change Order.
- F. Other Documents: Maintain manufacturer's certification, inspection certifications, field test records, and training documents required by individual Specification Sections.

1.4 SUBMITTALS

- A. No later than 90 days after Substantial Completion, deliver Record Documents and samples under provision of Section 01 77 00 PROJECT CLOSEOUT.
- B. Provide Portable Document Format (PDF) file of all record drawings on approved electronic media with record drawings (.dwg, .pdf, and .rvt).

PART 2 - PRODUCTS

NOT USED

PART 3 - EXECUTION

NOT USED

END OF SECTION 01 78 29

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SECTION 08 88 58 – ATCT CAB GLASS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Glass Products.
2. Insulating Glass.
3. Glazing Sealants.
4. Glazing Tapes.
5. Miscellaneous Glazing Materials.

1.2 DEFINITIONS

- A. Fenestration and Glazing Industry Alliance (FGIA): Glazing industry organization consisting of the consolidation of the former American Architectural Manufacturers Association (AAMA) and the former Insulating Glass Manufacturers Alliance (IGMA).
- B. Glass Manufacturers: Firms that produce primary glass, fabricated glass, or both, as defined in referenced glazing publications.
- C. Glass Fabricators: Firms that cut, laminate, temper, apply frit and produce Laminated and Insulating Glass Units (IGUs) and other related fabrication methods from stock glass manufactured by others.
- D. Glass Thicknesses: Indicated by thickness designations in inches or millimeters in accordance with ASTM C1036.
- E. IBC: International Building Code.
- F. Interspace: Space between lites of an insulating-glass unit.
- G. High Rise Building: A building when there is an occupied floor more than 75 feet above the lowest level of fire department vehicle access.
- H. Deterioration of Insulating Glass: Failure of the hermetic seal under normal use due to causes other than glass breakage and improper practices for maintaining, and cleaning insulating glass. Evidence of failure is the obstruction of vision by dust, moisture, or film on the interior surfaces of glass. Improper practices for maintaining and cleaning glass are those that do not comply with the manufacturer's directions.
- I. Outdoor/Indoor Transmission Class (OITC): OITC is a number rating of the sound transmission loss of a constructed assembly tested with lower frequencies.

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1.3 COORDINATION

- A. Coordinate glazing channel dimensions to provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances to achieve proper safety margins for glazing retention under each design load case, load case combination, and service condition.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site. Mandatory Attendees: FAA RE, General Contractor, Glazing Contractor.
 - 1. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
 - 2. Review temporary protection requirements for glazing during and after installation.
 - 3. Review crane, rigging, and other safety requirements.
 - 4. Review quality standards, inspection, and field-testing requirements.

1.5 ACTION SUBMITTALS

- A. Product Data:
 - 1. Float Glass.
 - 2. Spacers.
 - 3. Fasteners.
 - 4. Sealants.
 - 5. Glazing Tape.
 - 6. Setting and Edge Blocks.
- B. Shop Drawings:
 - 1. Include plans, elevations, and sections signed and sealed by licensed professional engineer.
 - 2. Include fastening spacing and details, frame type and details, setting and edge blocking, glass type(s) and thicknesses, glass clearances and bite and other glazing details.
- C. Glass Samples: Provide three 12-inch square samples. Sample must be indicative of finished glass product, including, but not limited to, edge finish.
- D. Glazing Accessory Samples: For sealants and spacers, in 12-inch lengths. Install sealant Samples between two strips of material representative in color of adjoining framing system.
- E. Glazing Schedule: List glass types and thicknesses for each size opening and location. Use same designations indicated on drawings.
- F. Delegated Design Submittals:

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1. Drawings: Provide complete design with plans, sections and details of glass, framing members, fastening, setting blocks, sealants and other applicable details.
2. Glass and Framing Calculations: Submit structural calculations signed and sealed by professional engineering in the state of Texas for all glass and framing members including the number and type of fasteners, sealant adhesions and other applicable items certifying compliance with performance requirements.
3. Seismic Design: Submit signed and sealed structural calculations and/or proof of testing certifying that the cab glass and framing and connections to primary structural systems and other building components accommodate the seismic relative displacements during MCE_R such that the cab glass must remain un-fractured and will not be dislodged from its framing under specified performance requirements.

1.6 INFORMATIONAL SUBMITTALS

- A. Qualification Data: Submit proof of qualifications for:
1. Testing and Laboratory Agencies.
 2. Manufacturer of Fabricated Glass.
 3. Installation Contractor.
 4. Delegated Design Professional.
- B. Product Certificates: For float glass.
- C. Product Test Reports: For fabricated glass and glazing sealants for tests performed by a qualified testing agency.
1. For glazing sealants, provide test reports based on testing current sealant formulations within previous 36-month period.
- D. Preconstruction adhesion and compatibility test report.
- E. Sample Warranties: For special warranties.

1.7 QUALITY ASSURANCE

- A. Delegated Design Professional: A qualified licensed professional engineer a minimum of ten (10) years' experience in high-rise glass and façade engineering and/or has successfully designed a minimum of three (3) complete ATCT cab-glazing systems.
- B. Fabricated-Glass Manufacturer Qualifications: A qualified manufacturer of fabricated glass for this Project who has a minimum of ten (10) years of experience fabricating glass products of similar type and size.
- C. Installer Qualifications: A qualified glazing contractor for this Project who has a minimum of five (5) years of glazing experience similar in material, design and extent to that indicated for Project with a record of successful in-service performance. Minimum experience and certification:

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1. The successful completion of at least 4,000 square feet of glazing for high-rise or ATCT construction.
 2. Project foreman must be certified under the National Glass Associations (NGA) Certified Glass Installer Program.
- D. Glass Testing Agency Qualifications: A qualified independent testing agency accredited according to the NFRC CAP 1 and 701 Certification Agency Program.
- E. Sealant Testing Agency Qualifications: An independent testing agency qualified according to ASTM C1021 to conduct the testing indicated.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Protect glazing materials in accordance with manufacturer's written instructions. Prevent damage to glass and glazing materials from adjacent construction activities, condensation, temperature changes, direct exposure to sun, or other causes.
- B. Comply with insulating-glass manufacturer's written instructions for venting and sealing units to avoid hermetic seal ruptures due to altitude change.
- C. Prior to delivery, all the cab glass shall be boxed and crated for shipping. Under no circumstances shall the cab glass be installed in the cab framing before delivering to the site. The cab glass shall be shipped to the site in a vertical position and stored in a vertical position against a sturdy support at an angle of approximately 7 degrees from vertical.
- D. Deliver products to the site in unopened containers, labeled plainly with manufacturer's names and brands.
- E. Store glass and setting materials in safe, dry locations with adequate ventilation free from heavy dust, not subject to combustion products or sources of water and that shall permit easy access for inspection and handling. Do not unpack until needed for installation.
- F. Deliver sealant compounds to the jobsite in sealed containers labeled to show the designated name, formula, or specification number; lot number, color, date of manufacturer, shelf life, and curing time when applicable.
- G. Handle and install materials in a manner that will protect them from damage.
- H. Unpack glass from the front of the case or container and avoid sliding the glass against itself or any uncushioned materials. Stack individual lites on edge using clean, cushioned pads placed at the quarter points of the bottom edge. Protect all edges from impact and use a clean dry separating material.

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1.9 FIELD CONDITIONS

- A. Environmental Limitations: Do not proceed with glazing when ambient and substrate temperature conditions are outside limits permitted by glazing material manufacturers and when glazing channel substrates are wet from rain, frost, condensation, or other causes.
 - 1. Do not install glazing sealants when ambient and substrate temperature conditions are outside limits permitted by sealant manufacturer or are below 40 degrees F unless lower temperatures are recommended by the glass and sealant manufactures and approved by the COR.
 - 2. Coordinate wind conditions with airport personnel, glazing contractor and crane operator.

1.10 WARRANTIES

- A. Manufacturer's Special Warranty for Insulating Glass: Manufacturer agrees to replace insulating-glass units that deteriorate within specified warranty period. Deterioration of insulating glass is defined as failure of hermetic seal under normal use that is not attributed to glass breakage or to maintaining and cleaning insulating glass contrary to manufacturer's written instructions. Evidence of failure is obstruction of vision by dust, moisture, or film on interior surfaces of glass.
 - 1. Warranty Period: Ten (10) years from date of CAI.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Source Limitations for Glass: Obtain glass from single source from single manufacturer.
- B. Source Limitations for Glazing Accessories: For each product and installation method, obtain from single source from single manufacturer.

2.2 GENERAL GLAZING SYSTEM

- A. Exterior Glazed: System must provide the capability to remove and re-install glass lites from the exterior of the building without requiring the removal or installation of fasteners or modifications to the framing system or the primary sealing system from the interior side of the building.

2.3 PERFORMANCE REQUIREMENTS

- A. General: Installed glazing systems shall withstand normal thermal movement, wind, and seismic loads without failure, including loss or glass breakage attributable to defective

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manufacture, fabrication, or installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; or other defects in construction.

- B. Delegated Design: The complete glazing system, including but not limited to, glass, framing, fasteners and all ancillary items must be designed by the delegated design professional.
- C. Structural Performance: Glazing shall withstand the following design loads within limits and under conditions indicated and in accordance with the IBC and ASTM E2461. The windload shall be derived from ASCE 7 (components and cladding) or from wind tunnel testing with loads indicated on the drawings. The design must accommodate the seismic relative displacements during MCE_R such that the cab glass must remain un-fractured and will not be dislodged from its framing under these displacements.
 - 1. Design Wind Pressures: As indicated on the Drawings
 - 2. Glass Probability of Breakage: Design for a probability of breakage not greater than 0.001.
 - 3. Maximum Lateral Glass Deflection: Limit center-of-glass deflection at design wind pressure to not more than 1/175 times the short-side length or 3/4 inch, whichever is less.
 - 4. Design Seismic Loads and Drift: As indicated on drawings.
 - 5. Deflection of Framing Members Supporting Glass: At design wind load, as follows:
 - a. Maximum Deflection Normal to Wall Plane: Limited to L/175 of clear span for spans of <
 - b. Deflection Parallel to Glazing Plane: Limited to L/175 or amount not exceeding that which reduces glazing bite to less than 75 percent of design dimension and that which reduces edge clearance between framing members and glazing or other fixed components to less than 1/8 inch (3.2 mm), whichever is less.
 - 6. Thermal Loads: Design glazing to resist thermal stress breakage induced by differential temperature conditions and limited air circulation within individual glass lites and insulated glazing units.
 - 7. Thermal and Optical Performance Properties: Provide glass with performance properties specified, as indicated in manufacturer's published test data, based on procedures indicated below:
 - a. For insulating-glass units, properties are based on units of thickness indicated for overall unit and for each lite.
 - b. U-Factors: Center-of-glazing values, in accordance with NFRC 100 and based on most current non-beta version of LBL's WINDOW computer program, expressed as Btu/sq. ft. x h x degrees F.
 - c. SHGC and Visible Transmittance: Center-of-glazing values, in accordance with NFRC 200 and based on most current non-beta version of LBL's WINDOW computer program.
 - d. Visible Reflectance: Center-of-glazing values, in accordance with NFRC 300.
- D. Acoustic Performance:
 - 1. Exterior Glazing: Minimum 34 OITC.

2.4 GLASS PRODUCTS, GENERAL

- A. Glazing Publications: Comply with the latest edition of published recommendations of glass product manufacturers and organizations below unless more stringent requirements are indicated. See these publications for glazing terms not otherwise defined in this Section or in referenced standards.
1. NGA Publications: “Laminated Glazing Reference Manual” and “GANA Glazing Manual.”
 2. AAMA Publications:
 - a. AAMA 101, North American Fenestration Standard/Specification for Windows, Doors, and Skylights.
 - b. AAMA 501.1, Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure.
 - c. AAMA 501.4, Recommended Static Test Method for Evaluating Window Wall, Curtain Wall and Storefront Systems Subjected to Seismic and Wind-Induced Inter-Story Drift.
 - d. AAMA 501.5, Test Method for Thermal Cycling of Exterior Walls.
 - e. AAMA 501.7, Recommended Static Test Method for Evaluating Windows, Window Wall, Curtain Wall and Storefront Systems Subjected to Vertical Inter-Story Movements.
 - f. AAMA 800, Voluntary Specification and Test Methods for Sealants.
 - g. AAMA 810.1, Voluntary Specification for Expanded Cellular Glazing Tape.
 - h. AAMA GDSG-1, Glass Design for Sloped Glazing.
 3. IGMA Publications:
 - a. IGMA TM-3000, North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial and Residential Use.
 - b. IGMA TB-3001, Guidelines for Sloped Glazing.
 4. NFRC:
 - a. NFRC 300 – Test Method for Determining the Solar Optical Performance of Glazing Materials and Systems.
 5. American Society of Testing and Materials (ASTM)
 - a. ASTM C1036 Standard Specification for Flat Glass
 - b. ASTM C1076 Standard Test Method for determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems.
 - c. ASTM C1172 Standard Specification for Laminated Architectural Flat Glass.
 - d. ASTM C1184 Standard Specification for Structural Silicone Sealants.
 - e. ASTM C1401 Standard Guide for Structural Sealant Glazing.
 - f. ASTM C864 Standard Specifications for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers.
 - g. ASTM D395 Standard Test Methods for Rubber Property-Compression Set.
 - h. ASTM E283 Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen.

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- i. ASTM E313 Standard Practice for Calculating Yellowness and Whiteness Indices from Instrumentally Measured Color Coordinates.
- j. ASTM E331 Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference.
- k. ASTM E547 Standard Test Method for Water Penetration of Exterior Windows, Skylights, and Curtain Walls by Cyclic Static Air Pressure Difference.
- l. ASTM E773 Standard Test Method for Accelerated Weathering of Sealed Insulating Glass Units.
- m. ASTM E783 Standard Test Method for Field Measurement of Air Leakage through Installed Exterior Windows and Doors.
- n. ASTM E1105 Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference.
- o. ASTM E1300 Standard Practice for Determining Load Resistance of Glass in Buildings.
- p. ASTM E1886 Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials.
- q. ASTM E1996 Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes.
- r. ASTM E2461 Standard practice for Determining the Thickness of Glass in Airport Traffic Control Tower Cabs.
- s. ASTM F2248 Standard Practice for Specifying an Equivalent 3-Second Duration Design Loading for Blast Resistant Glazing Fabricated with Laminated Glass.

B. Thickness: Where glass thickness is indicated, it is a minimum. Provide glass that complies with performance requirements and is not less than the thickness indicated.

C. Strength: Provide annealed float glass.

2.5 GLASS PRODUCTS

A. Low-Iron Annealed Float Glass:

1. Minimum thickness: 1/4 inch (6 mm). Match Existing glass size of IGU to be replaced.
2. Comply with ASTM C1036, Type I, Class I (clear), Quality-Q3 with the following exceptions:

Allowable Point Blemish Size and Distribution for lites $\geq 1/2$ inch (12 mm) ^A	
Blemish Size mm (inch)	Distribution
<1.80 <(0.07)	Allowed
≥ 1.80 <4.00 $\geq(0.07)$ <(0.16)	Allowed with a minimum separation of 600 mm (24 inch) ^B
≥ 4.00 $\geq(0.16)$	None allowed

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^A Table values are for thicknesses of 12.0 mm (1/2 inch) and greater. For glass less than 12.0 mm (1/2 in) thick, the provisions of ASTM C1036 apply.

^B Blemish Distribution: Blemishes in size ranges that are allowed without separation distance criteria shall not be compared to those with separation criteria. For example, there is no separation criteria between a blemish <1.8 mm and a blemish $\geq 1.80 < 4.00$ mm.

3. Optical Characteristics:
 - a. Visible Light Transmission: Greater than or equal to 91 percent. Match existing glass of existing IGU to be replaced.
 - b. Visible Light Reflectance: Less than or equal to 8 percent. Match existing glass of existing IGU to be replaced.
 - c. Match existing glass coloring of IGU to be replaced.
4. Known acceptable sources:
 - a. Vitro Architectural Glass Starphire Ultra-Clear[®] Low-Iron Float Glass.
 - b. Pilkington Optiwhite[™] Low Iron Float Glass.
 - c. Guardian UltraClear[®] Low-Iron Float Glass.
 - d. Saint Gobain Diamant[®] Low-Iron Float Glass.

2.6 INSULATING GLASS

- A. Insulating-Glass Units: Factory-assembled units consisting of sealed lites of glass separated by a dehydrated interspace.
 1. Sealing System: Dual seal, with polyisobutylene primary and silicone secondary sealants.
 2. Perimeter Spacer: Black, stainless steel fully filled desiccant. Spacers must be full length without joints along all edges. Joints must be connected with manufacturers approved corner key locks. Desiccant: Molecular sieve.
 3. Capillary Tubes: Use capillary tubes when recommended by glass manufacturer and fabricator based on elevation differences from fabrication site to Project site including methods and elevation of shipping route(s).
 - a. If recommended by glass manufacturer and fabricator, install in accordance with NGA/IGMA TB-1601-95(14) or as recommended by manufacturer/fabricator.
- B. Quality Standards:
 1. Qualified in accordance with ASTM E2190.
 2. Per ASTM C1036 with the following exceptions:
 - a. Maximum allowable point blemish size and distribution from interior to exterior lites as follows:

Insulating Glass Units and Insulating Laminated Glass Assemblies	
Allowable Point Blemish Size and Distribution ^A	
Blemish Size mm (inch)	Q3- Quality 3
<1.80 <(0.07)	Allowed

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$\geq 1.80 < 4.00$ $\geq (0.07) < (0.16)$	Allowed with a minimum separation of 600 mm (24 inch) ^B
≥ 4.00 $\geq (0.16)$	None allowed

^A Table applies to point blemishes from the exterior to interior lites of IGUs. The requirements of ASTM C1036 apply for each individual lite/layer.

^B Blemish Distribution: Blemishes in size ranges that are allowed without separation distance criteria shall not be compared to those with separation criteria. For example, for a double insulating unit with 1/4 inch lites there is no separation criteria when there is a blemish < 1.2 mm in exterior lite and a blemish $\geq 1.20 < 1.50$ mm in the interior lite.

2.7 GLAZING SEALANTS

A. General:

1. Compatibility: Compatible with one another and with other materials they contact, including glass products, seals of insulating-glass units, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
3. Colors of Exposed Glazing Sealants: As selected by COR from manufacturer's full range of industry colors.

B. Neutral-Curing Silicone Glazing Sealant, Class 50: Complying with ASTM C920, Type S, Grade NS, Use NT.

1. Known Acceptable Source:
 - a. DOWSIL™ 795 Silicone Building Sealant.

C. Glazing Tapes

D. Expanded Cellular Glazing Tapes: Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800.

1. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of wet sealant.

2.8 MISCELLANEOUS GLAZING MATERIALS

A. General: Provide products of material, size, and shape complying with referenced glazing standard, recommended in writing by manufacturers of glass and other glazing materials for application indicated, and with a proven record of compatibility with surfaces contacted in installation.

B. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

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C. Setting Blocks:

1. Silicone with Shore A durometer hardness of $85, \pm 5$.
2. Type recommended in writing by sealant or glass manufacturer.
3. Match existing setting block size and locations.

D. Spacers:

1. Neoprene blocks or continuous extrusions of hardness required by glass manufacturer to maintain glass lites in place for installation indicated.
2. Type recommended in writing by sealant or glass manufacturer.
3. Match existing spacer size and locations.

E. Edge Blocks:

1. Silicone: Shore-A Durometer hardness of $60, \pm 5$ hardness.
2. Type recommended in writing by sealant or glass manufacturer.
3. Match existing edge block size and locations.

F. Exposed Fasteners: Match existing fastener material. Field verify 316 stainless steel, UNC 2A.

2.9 FABRICATION OF GLAZING UNITS

A. Fabricate glazing units in sizes required to fit openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with written instructions of product manufacturer and referenced glazing publications, to comply with system performance requirements.

1. Allow for thermal movements from ambient and surface temperature changes acting on glass framing members and glazing components.
 - a. Temperature Change: 120 degrees F, ambient; 180 degrees F, material surfaces.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine framing, glazing channels, and stops, with Installer present, for compliance with the following:

1. Manufacturing and installation tolerances, including those for size, squareness, and offsets at corners.
2. Presence and functioning of weep systems.
3. Minimum required face and edge clearances.
4. Effective sealing between joints of glass-framing members.

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5. Glass color matching existing IGU to be replaced.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Each window must be measured individually. Measure glass for fabrication after frames have been installed and full dead loads of the cab floor and roof, along with any roof mounted equipment and where applicable the penthouse, have been applied.

B. Clean glazing channels and other framing members receiving glass immediately before glazing. Remove coatings not firmly bonded to substrates.

C. Examine glazing units to locate exterior and interior surfaces. Label or mark units as needed so that exterior and interior surfaces are readily identifiable. Do not use materials that leave visible marks in the completed work.

D. Leave removable labels in place until the installation is approved.

3.3 GLAZING, GENERAL

A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.

B. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site in accordance with Construction Waste and Management Specification. Damaged glass includes glass with edge damage or other imperfections that, when installed, could weaken glass, impair performance, or impair appearance.

C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.

D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.

E. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.

F. Provide spacers for glass lites where length plus width is larger than 50 inches.

1. Locate spacers directly opposite each other on both inside and outside faces of glass. Install correct size and spacing to preserve required face clearances, unless gaskets and glazing tapes are used that have demonstrated ability to maintain required face clearances and to comply with system performance requirements.

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2. Provide 1/8-inch minimum bite of spacers on glass and use thickness equal to sealant width. With glazing tape, use thickness slightly less than final compressed thickness of tape.
- G. Provide edge blocking where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and in accordance with requirements in referenced glazing publications.
- H. Use vacuum lifters rated for the weight of the glass with required safety factors to lift glass in place. Use suction cups to maneuver. Do not raise or drift glass with pry bars or similar devices.
- I. Set glass lites in each series with uniform pattern, draw, bow, and similar characteristics.

3.4 TAPE GLAZING

- A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.
- B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.
- C. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.
- D. Do not remove release paper from tape until right before each glazing unit is installed.
- E. Where indicated on drawings, apply heel bead of elastomeric sealant.
- F. Center glass lites in openings on setting blocks, and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.
- G. Apply cap bead of elastomeric sealant over exposed edge of tape.

3.5 GASKET GLAZING (DRY)

- A. Cut compression gaskets to lengths recommended by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.
- B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.
- C. Installation with Drive-in Wedge Gaskets: Center glass lites in openings on setting blocks, and press firmly against soft compression gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weathertight seal without

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developing bending stresses in glass. Seal gasket joints with sealant recommended in writing by gasket manufacturer.

- D. Installation with Pressure-Glazing Stops: Center glass lites in openings on setting blocks, and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended in writing by gasket manufacturer.
- E. Install gaskets so they protrude past face of glazing stops.

3.6 SEALANT GLAZING (WET)

- A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.
- B. Where indicated on the drawings, install glazing tape recessed in glazing channel indicated.
- C. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.
- D. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

3.7 GLASS INSPECTIONS

- A. Perform the following:
 - 1. Detailed inspection of each unit at fabrication plant for strict compliance with applicable standards. Ship only compliant units to site.
 - 2. Inspect for shipping damage after uncrating and prior to installing and setting the unit(s). Do not install any non-complaint units without prior written approval from the COR.
 - 3. Participate and provide necessary support to the FAA for final inspection of the completed product during the CAI activities of the Project.

3.8 CLEANING AND PROTECTION

- A. Immediately after installation, remove nonpermanent labels and clean surfaces in accordance with GANA Glazing Manual.
- B. Protect glass from contact with contaminating substances resulting from construction operations. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.

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1. If, despite such protection, contaminating substances do contact with glass, remove substances immediately as recommended in writing by glass manufacturer. Remove and replace glass that cannot be cleaned without damage.
- C. Remove and replace glass that is damaged during construction period.
- D. Wash glass on both exposed surfaces not more than four days before date scheduled for inspections that establish date of Substantial Completion. Wash glass in accordance with GANA Glazing Manual and as recommended in writing by glass manufacturer. Razor blades or other sharp objects shall not be used to clean glass surfaces.

3.9 INSULATING GLASS SCHEDULE

- A. Cab Glass: Low-Iron Insulating Glass
 1. Overall Unit Thickness: Match Existing IGU to be replaced.
 2. Minimum Thickness of Each Glass Lite: 1/4 inch (6 mm). Match Existing IGU to be replaced..
 3. Outdoor Lite: Low-iron annealed float glass.
 4. Interspace Content: Air.
 5. Indoor Lite: Low-iron annealed float glass.
 6. Match existing IGU glass coloring and characteristics.

3.10 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Field Quality-Control Testing: Perform the following test on representative areas of glazing.
 1. Water-Spray Test: Before installation of interior finishes has begun, areas designated by COR shall be tested in accordance with AAMA 501.2 and shall not evidence water penetration.
 - a. Perform a minimum of two tests in areas as directed by COR.
 2. Water Penetration: ASTM E1105 at a minimum cyclic static-air-pressure differential of 0.67 times the static-air-pressure differential specified for laboratory testing in "Performance Requirements" Article, but not less than 6.24 lbf/sq. ft. (300 Pa), and shall not evidence water penetration.
- C. Cab glass units will be considered defective if they do not pass tests and inspections.
- D. Prepare test and inspection reports.

END OF SECTION 08 88 58